

CERTIFICAT



CERTIFICADO



СЕРТИФИКАТ



認證證書



CERTIFICATE



ZERTIFIKAT



Management Service

CERTIFICATE

Certificate Registration No.: **12 100 68743 TMS** / Order No.: **707190220**

**The Certification Body
of TÜV SÜD Management Service GmbH**

certifies that the organization



Burkert UK Limited

**1 Bridge End
Cirencester GL7 1QY
United Kingdom**

for the scope

**Suppliers of process equipment to
measure and control fluids, coordinating design and build**

has established and applies a Quality Management System.

An audit was performed and has furnished proof
that the requirements according to

DIN EN ISO 9001:2015

are fulfilled.

The certificate is valid from **2025-01-26** until **2028-01-25**.

Fred Wenke
Head of Certification Body
Munich, 2025-01-09



Typ FULLBORE S055/S051/S054

SENSORS

Installation manual

DE

Ausführliche Informationen finden Sie in der Bedienungsanleitung
unter der Internetadresse:

country.burkert.com > S055_S051_S054

oder

scannen Sie folgenden QR-Code ein:



EN

Detailed information can be found in the operating instructions
at the Internet address:

country.burkert.com > S055_S051_S054

or

scan the following QR code:



FR

Vous trouverez des informations détaillées dans le mode d'emploi
à l'adresse Internet suivante :

country.burkert.com > S055_S051_S054

ou

scannez le code QR suivant :



INDEX

INTRODUCTION	2
START UP AND MAINTENANCE OF THE INSTRUMENTS	2
SAFETY	3
GENERAL INFORMATION ON THE SENSORS INSTALLATION	3
ELECTRICAL CONNECTIONS OF SENSOR TO TRANSMITTER	6
GROUNDING INSTRUCTIONS	7
TORQUES (NM) FOR SENSOR BOLTS (FLANGED & WAFER)	8

INTRODUCTION

- This manual is integral part of the product. Read carefully the instructions contained since it contains important indications for the safety of use and of maintenance.
- The technical information and the relative products of this manual could be modified without any previous notice.
- The flowmeter must be used for the use it has been built for. The improper use, possible tampering of the instrument or parts of it and substitutions of any components not original, makes the warranty to decay automatically.
- The manufacturer is considered responsible only if the instrument is used in its original configuration and setting.
- The flowmeter makes measures of liquids with conductivity greater than $5\mu\text{S}/\text{cm}$; it consists of a sensor (described in this manual) and a converter, for it see the manual provided.
- If the sensor is supplied in compact version to the converter, consider the operating temperatures more restrictive, otherwise refer to the respective manuals.
- When transporting, unpacking and handling the flowmeter, be careful and care.
- In the case of prolonged storage and of transport, use and store in the original container in a dry place, do not place more than 3 packs one above the other. It is possible pallets storage and transport (in case of wooden crates do not place one above the other).
- For the cleaning of the device use only a damp cloth, and for the maintenance/repairs, contact the customer service.
- For the disposal of the device and of the packaging make strict reference to the regulations.
- It is forbidden the reproduction of the present manual and of possible software supplied with the instrument.

START UP AND MAINTENANCE OF THE INSTRUMENTS

- Before starting up the instrument, always make a secure connection to ground as suitable to page 6.
- Verify periodically: the cables integrity, the tightening of the sealing elements (cable glands, covers, etc.), the mechanical fixing of the instrument on the pipe or on the wall.

SAFETY



Before using the instrument, always make a secure connection to the ground



Avoid any attempt to repair the instrument. If the instrument is not functioning properly, please call the nearest assistance service



Pay maximum attention during the operations



ATTENTION !!



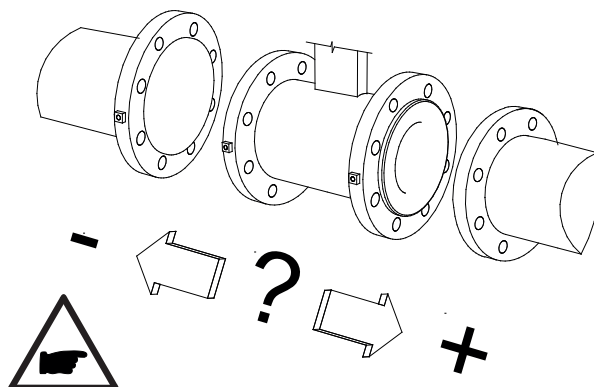
DANGER !!!

GENERAL INFORMATION ON THE SENSORS INSTALLATION

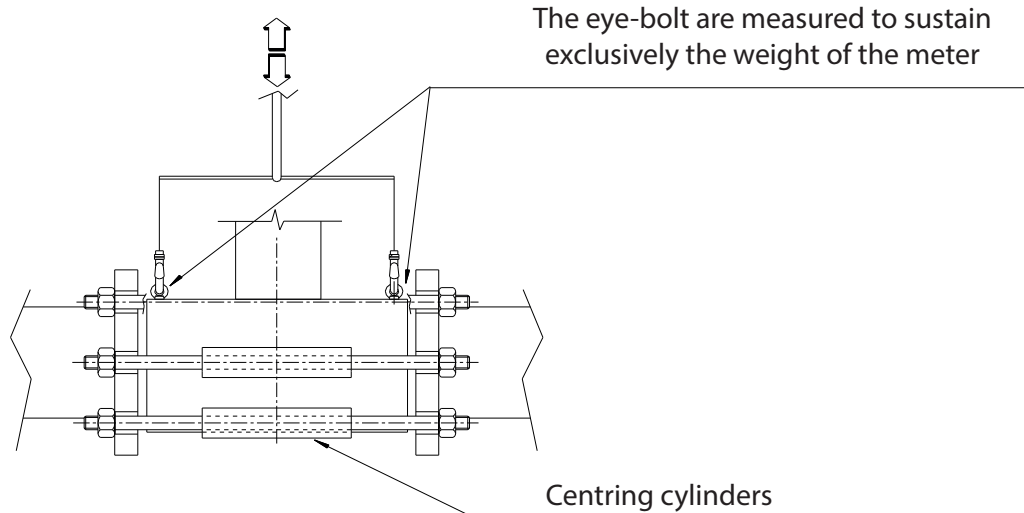
Flow direction

Before installing the sensor check the direction of the liquid in the piping.

The sign of the flow rate is positive, when the flow direction is from – to + as printed on the tag plate. If the device is mounted in the reverse normal flow direction, the sign of the flow rate can be corrected by changing the sign of the coefficient KA.



Installation (Method recommended for ALL THE SENSOR WITH EYEBOLT)

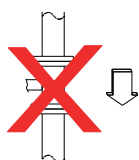


N.B.: For sensor S054 we recommend the use of centring cylinders

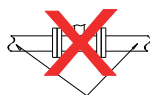
Shrewdness and precautions

NO

YES



For vertical installations with descending flow direction contact the manufacturer

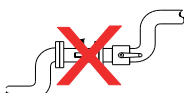


LONG PIPE

Avoid the installation of the sensor in a long pipe, without any support of the same



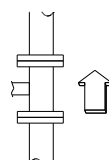
Avoid operating when the pipe is partially empty



Avoid the installation next to bows in pipes and respect min. inlet and outlet sections



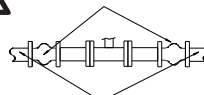
Avoid the approach of the flange and counter flange using the closing force of the nuts



For vertical installations is preferable an ascending flow

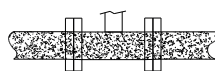


ANTI VIBRATION JOINTS

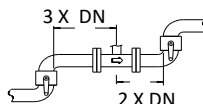


LONG PIPE

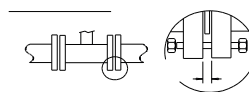
For installations on long pipes, please use the anti vibration joints



During operation, the pipe must be either completely full of liquid, or completely empty



Install the sensor at a min. distance to bows in pipes and hydraulic accessories



GASKET THICKNESS + 4mm

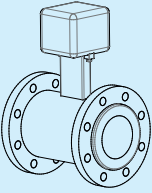
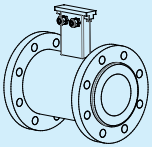
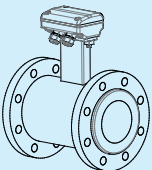
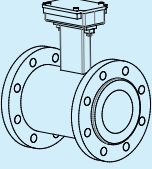
Before tightening the nuts, place both the flanges of pipe and device as near together as possible.

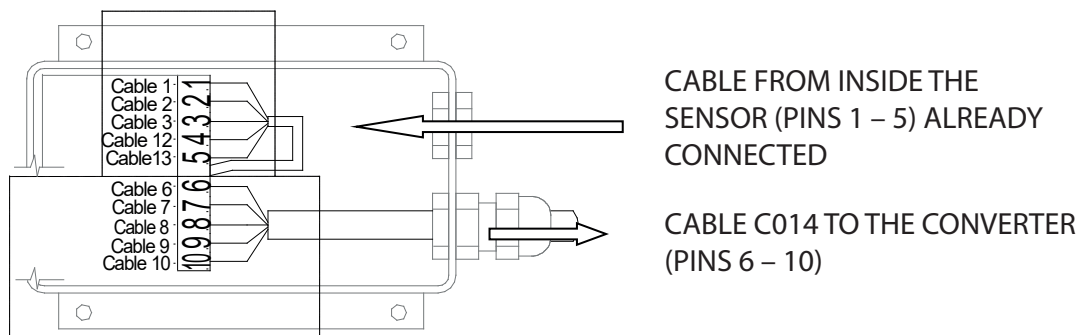
OPERATING TEMPERATURES

	EBONITE LINING				PP LINING				PTFE LINING			
	Liquid Temp.		Ambient Temp.		Liquid Temp.		Ambient Temp.		Liquid Temp.		Ambient Temp.	
	Min.	Max.	Min.	Max.	Min.	Max.	Min.	Max.	Min.	Max.	Min.	Max.
°C	0	80	-5	60	0	60	0	60	-20	130	-10	60
°F	30	176	23	140	32	140	32	140	-4	266	14	140

ELECTRICAL CONNECTIONS OF SENSOR TO TRANSMITTER

(Connections to transmitter: see related manual)

	VERSION	SUITABLE FOR	SENSOR'S CONNECTION
	COMPACT	ALL SENSORS MODEL	NO CONNECTIONS
	SEPARATE WITHOUT JUNCTIONS-BOX	STAINLESS STEEL MODEL	NO CONNECTIONS REQUIRED (CABLE ALREADY CONNECTED AND POTTED)
	SEPARATE WITH OR WITHOUT JUNCTIONS-BOX	ALL CARBON STEEL MODELS	NO CONNECTIONS REQUIRED (CABLE ALREADY CONNECTED AND POTTED)
	SEPARATE WITH PREAMPLIFIER	ALL SENSORS MODEL	SEE BELOW

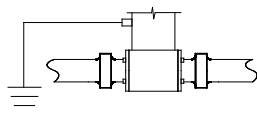


GROUNDING INSTRUCTIONS

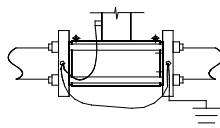


For correct operation of the meter, it is **NECESSARY** that the sensor and the liquid are equipotential, so **ALWAYS** connect the sensor and converter to ground:

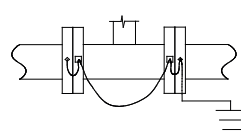
GROUNDING WITH METALLIC PIPE



Sensors with ground socket
on the connection box

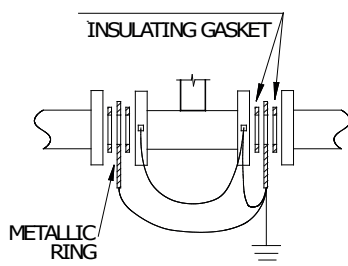


Wafer sensors



Flanged sensors

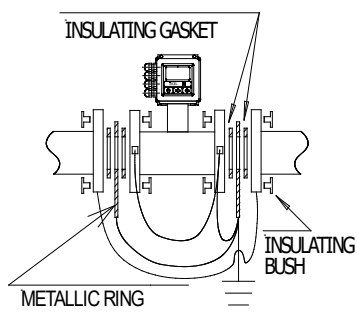
GROUNDING WITH INSULATING PIPE



If the sensor has to be mounted on a pipe made of an insulating material is necessary:

- Install two metallic ring between the sensor flanges and the counter flanges of the pipe line or:
- Use a sensor with the additional grounding electrode

GROUNDING WHEN THERE IS A CATHODIC PROTECTION OVER THE PIPE



If the sensor must be install in the piping with a cathode protection, is necessary:

- using insulating bushes to isolate the bolts
- Grounding metallic rings should be provided to ground the liquid using insulating gasket between the rings
- **IMPORTANT :**
The ripple of DC power source used for cathodic protection shall be = 0



TORQUES (NM) FOR SENSOR BOLTS (FLANGED & WAFER)

OPERATIVE PRESSURE										
Kpa	1000		1600			2500		4000		6400
psi	140		260			350		600		1000
DN	PTFE	EBON.	PTFE	EBON.	PP	PTFE	EBON.	PTFE	EBON.	EBON.
25			25 (21)		19	25		25 [32]		39 [32]
32			43 (26)		28	43		43 [40]		53 [40]
40			53 (32)		36	53		53 [63]		72 [63]
50			68 (60)		52	68		68 [35]		81 [35]
65			90 (78)		75	45		45 [53]		58 [53]
80			53 (89)		41	53		53 [68]		62 [68]
100			59 (70)		56	83		83 [94]		87 [94]
125			77 (94)		71	112		112 [130]		148 [130]
150			108 (106)		106	135		135 [113]		217 [113]
200	148	432	99 (148)	288 (433)		134	391	178 [178]	520 [519]	816 [519]
250	123	359	140 (156)	408 (455)		204	595	267 [185]	780 [540]	1124 [540]
300	142	415	175 (234)	510 (683)		201	588	278 [275]	812 [803]	1108 [803]
350	172	502	205 (325)	598 (946)		324	945	422 [318]	1231 [927]	1684 [927]
400	217	632	282 (317)	821 (911)		426	1243	619 [411]	1805 [1198]	2180 [1198]
450	194	564	281 (336)	981 (926)				[398]	[1161]	
500	224	652	382 (317)	1113 (924)				[465]	[1356]	
550			(379)	(1105)				[608]	[1772]	
600	323	942	568 (463)	1658 (1350)				[774]	[2258]	
650			(429)	(1251)				[753]	[2195]	
700	356	1040	421 (503)	1230 (1468)				[947]	[2761]	
750			(451)	(1315)				[1105]	[3223]	

800	476	1388	549	1603						
850			(563)	(1642)				[1373]	[4006]	
900	450	1312	519 (618)	1515 (1803)				[1408]	[4106]	
1000	582	1699	721 (736)	2105 (2146)				[1598]	[4662]	

- Tighten uniformly in an alternating pattern
- The torque listed in tab are applicable to flanges:
EN1092-1, DIN 2501, BS 4504, ANSI B16.5 , JIS
- Use gaskets acc. to DIN 2690
- For DN > 1000 contact the manufacturer
- (**)= ANSI 150
- [**]= ANSI 300

At the end of its lifetime, this product shall be disposed of in full compliance with the environmental regulations of the state in which it is located.

REVIEW	DATE	DESCRIPTION
MAN_FULLBORE_S055_S020_S054_EN_BU_R06	23/03/2021	GRAPHIC UPDATE AND CONTENT CORRECTIONS



2/2-way or 3/2-way ball valve, 2-piece

- Stainless steel ball valve
- High flow rate
- High medium pressure
- Long service life
- ISO 5211 mounting flange



Product variants described in the data sheet may differ from the product presentation and description.

Can be combined with



Type 2051
Pneumatic rotary actuator



Type 3003
Electrical Rotary Actuator
- On/Off and control



Type 3004
Explosion-Proof Rotary
Actuator - On/Off or Control



Type 3005
Electric Rotary Actuator - On/
Off and Control



Type 1061
Position feedback unit for
pneumatic rotary actuators



Type 8792
Digital electropneumatic
positioner SideCONTROL



Type 8805
Ball valve / Butterfly valve
with pneumatic rotary
actuator



Type 8804
Ball valve / butterfly valve
with electrical rotary actuator

Type description

2/2- and 3/2-way stainless steel ball valve used to shut off and distribute medium flows. The ball valves can be connected via the mechanical interface according to ISO 5211 with a pneumatic rotary actuator (e.g. Type 2051, Type 2052, Type 2053) or electrical rotary actuator (e.g. Type 3003, Type 3004, Type 3005).

Table of contents

1. General technical data	3
2. Approvals and conformities	3
2.1. General notes	3
2.2. Conformity	3
2.3. Standards	3
3. Materials	4
3.1. Bürkert resistApp	4
4. Dimensions	4
4.1. 2-way ball valve, threaded variant	4
4.2. 3-way ball valve, threaded variant	5
5. Performance specifications	5
5.1. Pressure temperature diagram	5
5.2. Torques	6
5.3. Switching positions 3-way ball valve	6
6. Ordering information	6
6.1. Bürkert eShop	6
6.2. Bürkert product filter	6
6.3. Ordering chart 2-way ball valve, threaded variant	7
6.4. Ordering chart 3-way ball valve, threaded variant	7

1. General technical data

Product properties	
Dimensions	Further information can be found in chapter "4. Dimensions" on page 4.
Materials	
Seal	PTFE (ball seal) FKM (stem sea)
Body	Stainless steel 1.4408/316
Ball	Stainless steel 1.4401/316
Stem	Stainless steel 1.4401/316
Nominal diameter	DN 10...50 (2/2-way) DN 10...40 (3/2-way)
Ball bore (3/2-way)	L-bore T-bore
Medium data	
Medium temperature	- 10 °C...+ 200 °C (see "5.1. Pressure temperature diagram" on page 5)
Medium pressure	2/2-way: max. 100 bar (see "5.1. Pressure temperature diagram" on page 5) 3/2-way: max. 63 bar (see "5.1. Pressure temperature diagram" on page 5)
Product connections and communication	
Port connection	Rp 1/4"...Rp 2" Whitworth thread according to DIN EN 10226 - 1 (old DIN 2999)
Actuator-side interface	According to EN ISO 5211

2. Approvals and conformities

2.1. General notes

- The approvals and conformities listed below must be stated when making enquiries. This is the only way to ensure that the product complies with all required specifications.
- Not all available variants can be supplied with the below mentioned approvals or conformities.

2.2. Conformity

In accordance with the Declaration of Conformity, the product is compliant with the EU Directives.

2.3. Standards

The applied standards which are used to demonstrate compliance with the EU Directives are listed in the EU-Type Examination Certificate and/or the EU Declaration of Conformity.

3. Materials

3.1. Bürkert resistApp



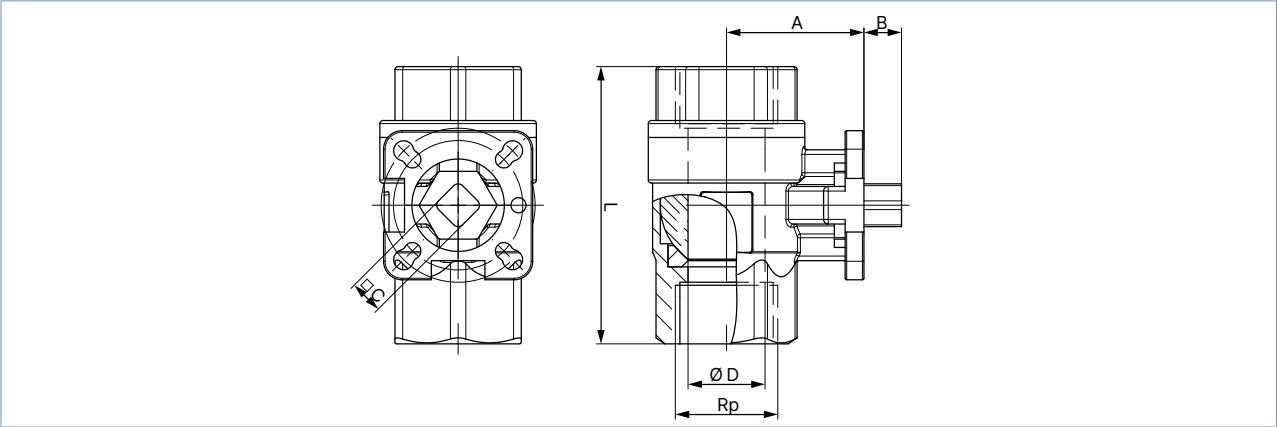
Bürkert resistApp – Chemical resistance chart

You want to ensure the reliability and durability of the materials in your individual application case? Verify your combination of media and materials on our website or in our resistApp.

Start chemical resistance check

4. Dimensions

4.1. 2-way ball valve, threaded variant



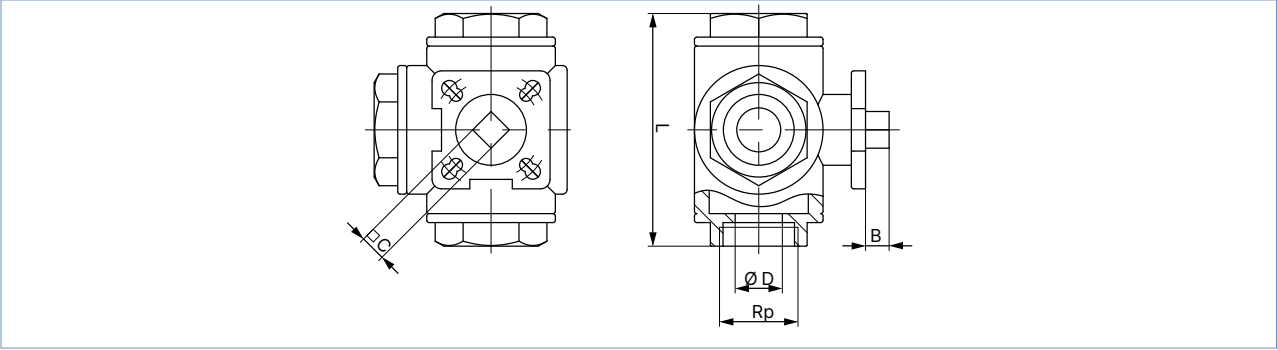
DN	Rp [inch]	D [mm]	L [mm]	A [mm]	B [mm]	C [mm]	Interface acc. to ISO 5211	Article no.	
								Bare shaft ¹⁾	Manually operated
10	¼	11.6	50	33.2	7	9	F03	185972	276475
12	⅜	12.7	60	33.2	7	9	F03/F04	185973	276476
15	½	15	75	35	7.5	9	F03/F04	185974	276477
20	¾	20	80	41	12.1	11	F04/F05	185975	276478
25	1	25	90	44.5	12.1	11	F04/F05	185976	276479
32	1¼	31.8	110	49.5	11.5	11	F04/F05	185977	276480
40	1½	38	120	55	12.1	11	F04/F05	185978	276481
50	2	50.8	140	66.5	15.5	14	F05/F07	185979	276482

1.) The ball valves with bare shaft are suitable for automated operation with pneumatic or electric rotary actuators.

4.2. 3-way ball valve, threaded variant

Note:

Further information on 3-way ball valves with hand lever can be found in **Type TKU001|002|003|004** ▶.

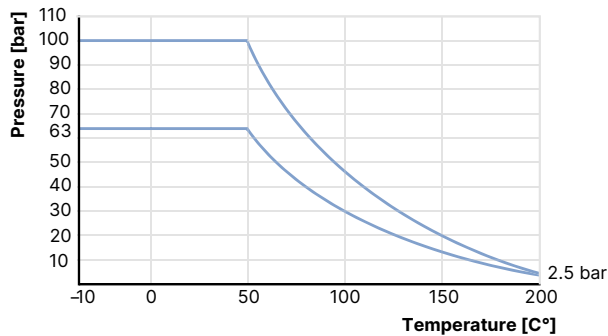


DN	Rp [inch]	D [mm]	L [mm]	B [mm]	C [mm]	Interface acc. to ISO 5211	Article no.	
							L-bore	T-bore
8	1/4	12	80	8.5	9	F03/F04	172844	185984
10	3/8	12	80	8.5	9	F03/F04	185981	185986
15	1/2	12	80	8.5	9	F03/F04	185853	185987
20	3/4	15	88	11.5	11	F04/F05	172845	185988
25	1	20	100	11.5	11	F04/F05	172847	185989
32	1 1/4	25	122	11.5	11	F04/F05	172848	185990
40	1 1/2	32	141	11.5	11	F04/F05	172849	185992
50	2	40	171	15.5	14	F05/F07	172851	185993

5. Performance specifications

5.1. Pressure temperature diagram

For general industrial use



5.2. Torques

Note:

The values were measured under the following conditions: Max. Δp , ambient temperature, medium: water

Torques for 2/2-way ball valve with a PN 100 pressure rating								
DN [mm]	10	12	15	20	25	32	40	50
Breakaway torque [Nm]	13	13	16	25	31	53	76	99
Running torque [Nm]	10	10	13	20	24	43	60	80

Torques for 3/2-way ball valve with a PN 63 pressure rating								
DN [mm]	10	12	15	20	25	32	40	
Breakaway torque [Nm]	13	13	18	22	35	42	68	
Running torque [Nm]	9	9	12	14	23	28	45	

5.3. Switching positions 3-way ball valve

Ball bore		T			L
Position	0°				
	90°				
Switching position		T1	T2	T3	L4

6. Ordering information

6.1. Bürkert eShop



Bürkert eShop – Easy ordering and quick delivery

You want to find your desired Bürkert product or spare part quickly and order directly? Our online shop is available for you 24/7. Sign up and enjoy all the benefits.

[Order online now](#)

6.2. Bürkert product filter



Bürkert product filter – Get quickly to the right product

You want to select products comfortably based on your technical requirements? Use the Bürkert product filter and find suitable articles for your application quickly and easily.

[Try out our product filter](#)

6.3. Ordering chart 2-way ball valve, threaded variant

Note:

Further information can be found in chapter [“4.1. 2-way ball valve, threaded variant” on page 4.](#)

6.4. Ordering chart 3-way ball valve, threaded variant

Note:

Further information can be found in chapter [“4.2. 3-way ball valve, threaded variant” on page 5.](#)

DATA SHEET

Type 8804



2/2 or 3/2 way ball valve with electrical rotary actuator

- High flow rate value
- Actuator with adjustable limit switches
- Visual position indicator
- Manual override in standard

Product variants described in the data sheet may differ from the product presentation and description.

Type description

The electrical ball valve Type 8804 consists of an electrical rotary actuator and a 2/2 or 3/2-way ball valve made of stainless steel. 2-way variants are available with a 2-piece or 3-piece ball valve for various applications. The 3-way ball valves are available in L and T bore version. The rotary actuator is compactly built. Moreover it can be used as single device for the operation of other control elements (see datasheet Type 3003). Heat resistors and torque limiters are standard. The body is made of low inflammable material, classified acc. to UL 94 V0.

Table of contents

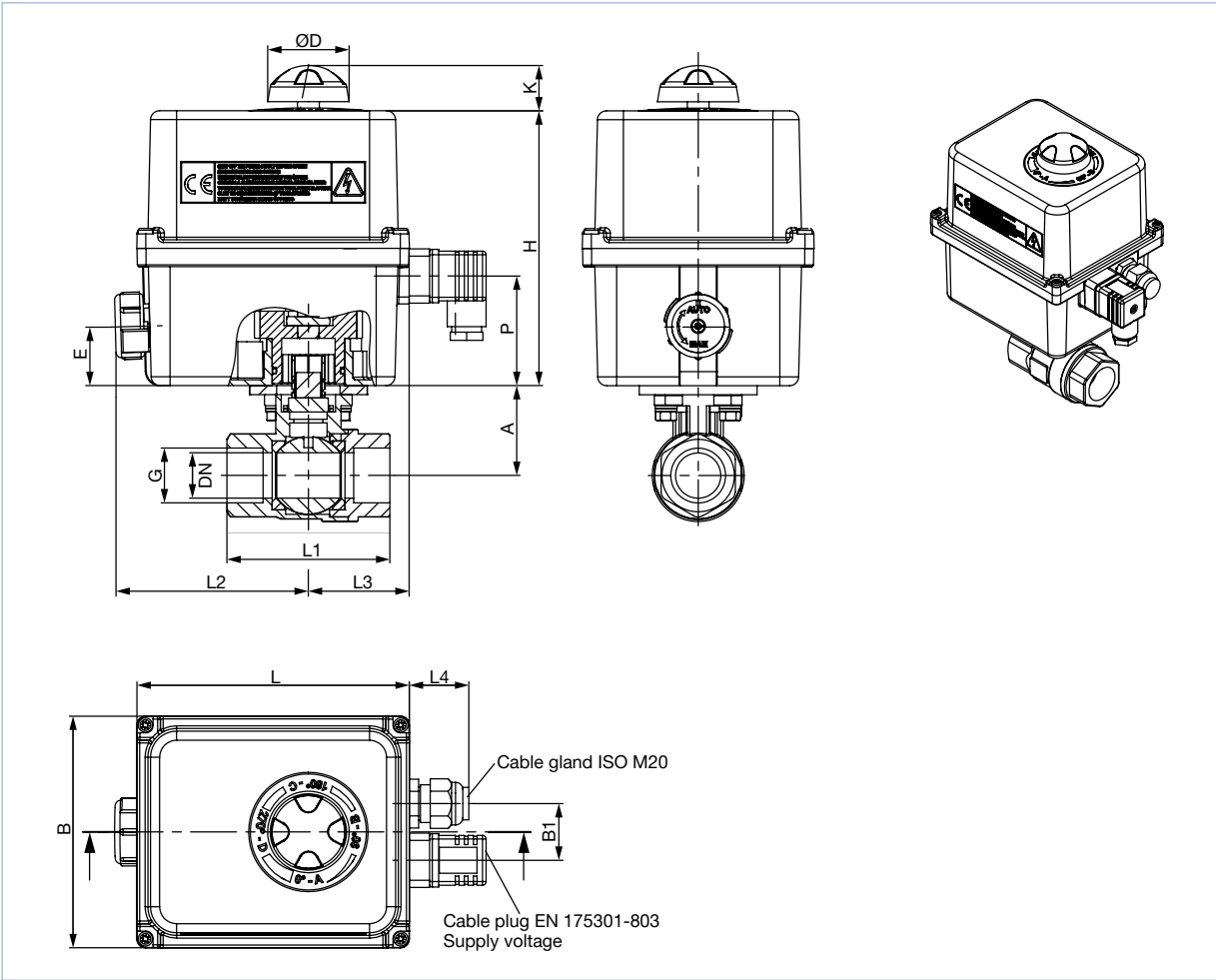
1. General technical data	3
2. Dimensions	4
2.1. 2-piece ball valve	4
2.2. 3-piece ball valve	5
3. Device / Process connections	6
3.1. Connection details	6
Version with analogue signal input	7
4. Performance specifications	8
4.1. Pressure temperature diagram for stainless steel ball valves	8
5. Ordering information	8
5.1. Bürkert eShop – Easy ordering and quick delivery	8
5.2. Bürkert product filter	8
5.3. Ordering chart	9
5.4. Ordering chart accessories	10
Conversion sleeves	10
Replacement battery for actuators with electronic emergency reset function	10

1. General technical data

Product characteristics	
Dimensions	Detailed information can be found in chapter “2. Dimensions” on page 4.
Material	
Body	Stainless steel 1.4408
Seal	PTFE
Material electrical actuator	
Cover/Body	Nylon/PA 6.6
Axis/Screws	Stainless steel
Gear	Stainless steel and PC
Connection between actuator and ball valve	Flange according to ISO 5211
Duty cycle	According to IEC34 S4 = 50 %
Actuating time	Detailed information can be found in chapter “5.3. Ordering chart” on page 9
Limit switches	4 adjustable (2 for the motor and 2 additional ones for feedback) Max. 250 V AC/5 A
Electrical data	
Operating voltage	15...30 V AC 50/60 Hz / 12...48 V DC 100...240 V AC 50/60 Hz / 100...350 V DC
Voltage tolerance	± 10 % At 12...48 V DC the operating voltage must not drop below 11.5 V
Media data	
Media	Gaseous and liquid media which do not attack the body and seal materials
Media temperature	Depending on the used valve and medium pressure, see “4.1. Pressure temperature diagram for stainless steel ball valves” on page 8. The temperature limit values of the actuators are usually lower than those of the valves. Therefore it has to be ensured, that the temperature limits of other system components (like actuators) do not exceed due to a possible heat radiation from the medium.
Process/Port connection & communication	
Port connection	Rp ¼" - Rp 2½" Whitworth Thread acc. to DIN EN 10226-1 (old DIN 2999) Weld end (on request)
Electrical connection	Cable plug acc. to EN175301-803 (supply voltage) included Cable glands ISO M20
Environment and installation	
Ambient temperature	
For Actuator 3003 Standard	- 10 °C...+ 55 °C
For Actuator 3003 with electronic emergency reset	- 10 °C...+ 40 °C
Rotation angle	90° (± 5°)
Installation position	Do not install actuator with the cover face down (upside down)!
Degree of protection	IP66 mounted with cable plug

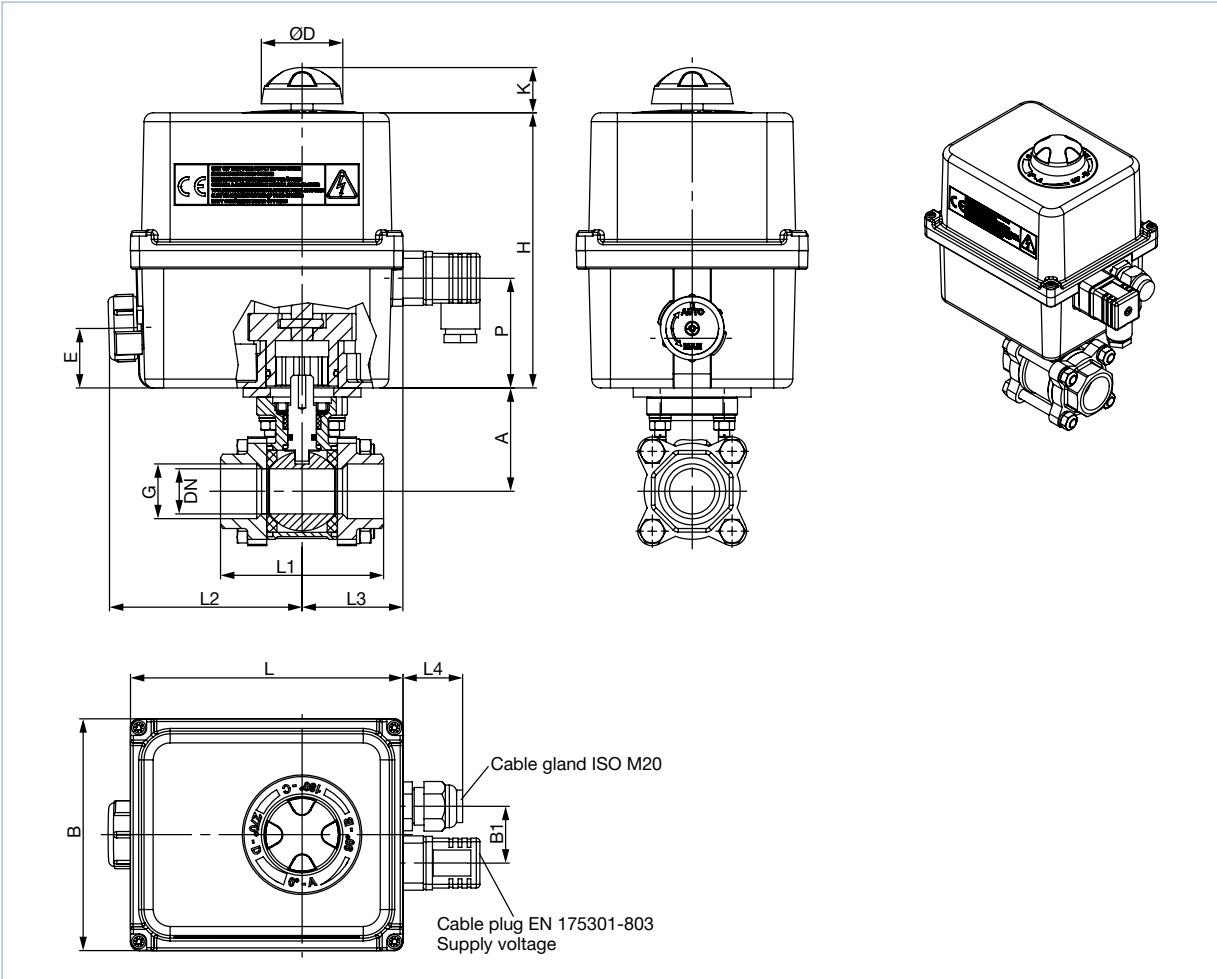
2. Dimensions

2.1. 2-piece ball valve



Actuator	DN	Port connection	A	B	B1	ØD	E	H	K	L	L1	L2	L3	L4	P
20 Nm	10	¼"	33.2	91.7	30.0	37.1	23.5	123.7	24.5	136.5	50.0	103.6	45.4	29.7	47.0
	13	⅜"	33.2								60.0				
	15	½"	35								75.0				
	20	¾"	41								80.0				
35...100 Nm	10	¼"	44.2	127.7	31.3	37.1	32.5	150.8	25.5	150.3	50.0	106.3	55.7	29.7	60.5
	13	⅜"	44.2								60.0				
	15	½"	46.0								75.0				
	20	¾"	41.0								80.0				
	25	1"	44.5								90.0				
	32	1¼"	49.5								110.0				
	40	1½"	55.0								120.0				
	50	2"	66.5								140.0				
	65	2½"	101.0								185.0				

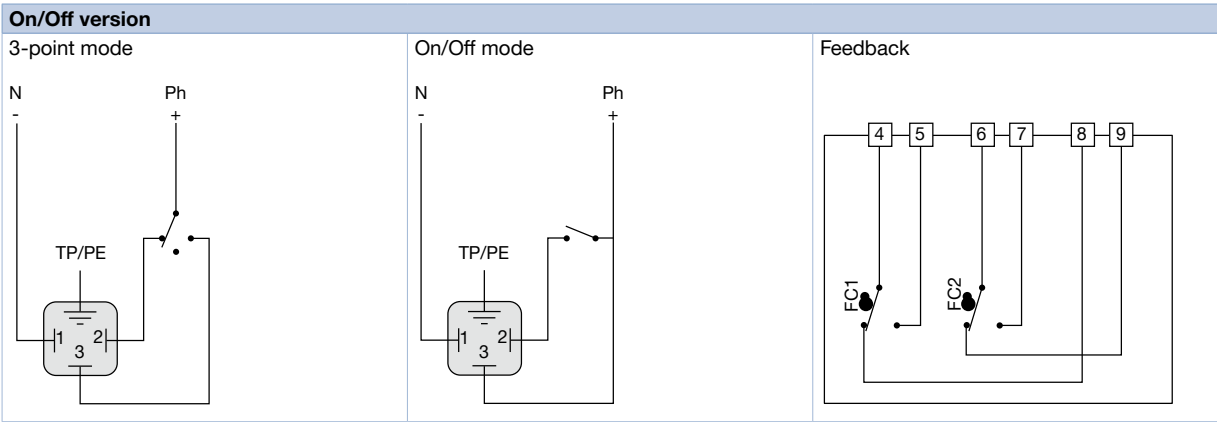
2.2. 3-piece ball valve



Actuator	DN	Port connection	A	B	B1	ØD	E	H	K	L	L1	L2	L3	L4	P
20 Nm	10	1/4"	40	91.7	30.0	37.1	23.5	123.7	24.5	136.5	65	103.6	45.4	29.7	47.0
	12	3/8"									75				
	15	1/2"									80				
35...100 Nm	20	3/4"	44	127.7	31.3	37.1	32.5	150.8	25.5	150.3	65	106.3	55.7	29.7	60.5
	10	1/4"	40								75				
	12	3/8"									80				
	15	1/2"									90				
	20	3/4"	44								110				
	25	1"	52								120				
	32	1 1/4"	58								140				
	40	1 1/2"	68								185				
	50	2"	77												
	65	2 1/2"	98												

3. Device / Process connections

3.1. Connection details

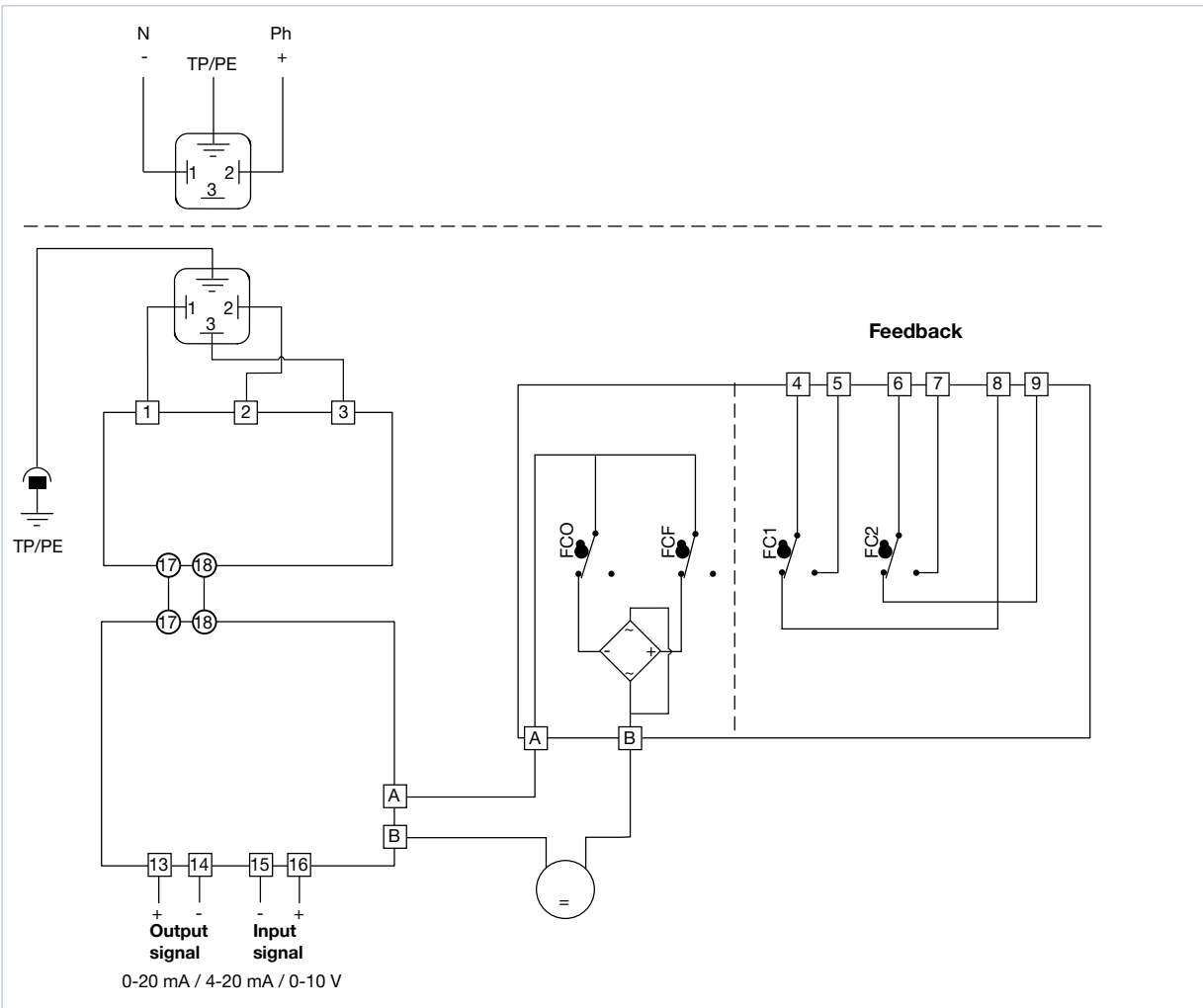


Symbol	Description
FCO	Limit switch OPEN
FCF	Limit switch CLOSED
FC1	Additional limit switch 1
FC2	Additional limit switch 2

Version with analogue signal input

Note:

- The resolution of the closed-loop control mode is 1°.
- The input resistance for control 0...10 V is 10 kΩ.
- The input resistance for control 0...20 mA/4...20 mA is 100 Ω.



Symbol	Description
FCO	Limit switch OPEN
FCF	Limit switch CLOSED
FC1	Additional limit switch 1
FC2	Additional limit switch 2

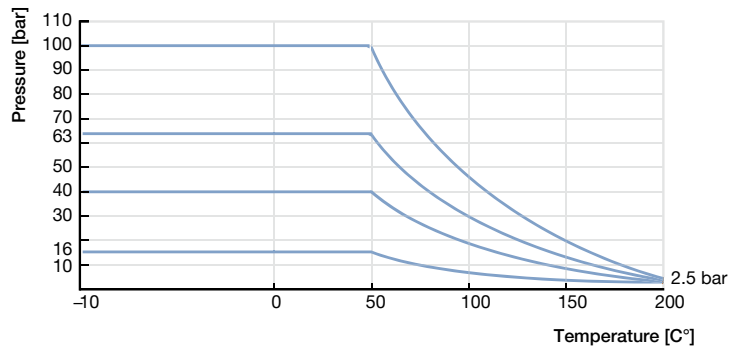
4. Performance specifications

4.1. Pressure temperature diagram for stainless steel ball valves

Note:

The temperature limits of the actuators are usually lower. Therefore the lowest temperature limit values of the included system components (ball valve and actuator) must be observed.

For general industrial use



5. Ordering information

5.1. Bürkert eShop – Easy ordering and quick delivery

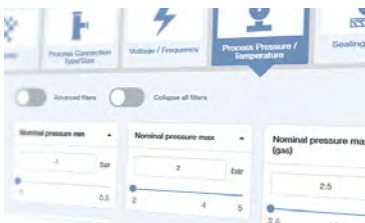


Bürkert eShop – Easy ordering and fast delivery

You want to find your desired Bürkert product or spare part quickly and order directly? Our online shop is available for you 24/7. Sign up and enjoy all the benefits.

[Order online now](#)

5.2. Bürkert product filter



Bürkert product filter – Get quickly to the right product

You want to select products comfortably based on your technical requirements? Use the Bürkert product filter and find suitable articles for your application quickly and easily.

[Try out our product filter](#)

5.3. Ordering chart

Nominal diameter	Port connection	Flow coefficient water	Pressure range	Actuating time for 90°	Actuator	Voltage ^{1.)}	Article no. 2-piece
[mm]		[m³/h]	[bar]	[s]	[Nm]		
10.0	¼"	7	0...100	012	020	100...240 V AC 50/60 Hz 100...350 V DC	226466 
12.7	⅜"	9	0...100	012	020		226468 
15.0	½"	19	0...100	012	020		226469 
20.0	¾"	46	0...100	007	035		226470 
25.0	1"	72	0...100	007	035		226471 
32.0	1¼"	105	0...100	012	060		226472 
40.0	1½"	170	0...100	023	100		226473 
10.0	¼"	7	0...100	012	020	15...30 V AC 50/60 Hz 12...48 V DC	226475 
12.7	⅜"	9	0...100	012	020		226476 
15.0	½"	19	0...100	012	020		226477 
20.0	¾"	46	0...100	007	035		226478 
25.0	1"	72	0...100	007	035		226479 
32.0	1¼"	105	0...100	012	060		226480 
40.0	1½"	170	0...100	023	100		226481 

1.) For 12...48 V DC the operating voltage should not go below 11.5 V.

Nominal diameter	Port connection	Flow coefficient water	Pressure range	Actuating time for 90°	Actuator	Voltage ^{1.)}	Article no. 3-piece
[mm]		[m³/h]	[bar]	[s]	[Nm]		
10.0	¼"	7	0...63	012	020	100...240 V AC 50/60 Hz 100...350 V DC	226483 
12.7	⅜"	9	0...63	012	020		226484 
15.0	½"	19	0...63	012	020		226485 
20.0	¾"	46	0...63	012	020		226486 
25.0	1"	72	0...63	007	035		226487 
32.0	1¼"	105	0...63	007	035		226488 
40.0	1½"	170	0...63	012	060		241107 
50.0	2"	275	0...63	012	060	15...30 V AC 50/60 Hz 12...48 V DC	226490 
10.0	¼"	7	0...63	012	020		226496 
12.7	⅜"	9	0...63	012	020		226497 
15.0	½"	19	0...63	012	020		226498 
20.0	¾"	46	0...63	012	020		226499 
25.0	1"	72	0...63	007	035		226500 
32.0	1¼"	105	0...63	007	035		225945 
40.0	1½"	170	0...63	012	060		241109 
50.0	2"	275	0...63	012	060		226502 

1.) For 12...48 V DC the operating voltage should not go below 11.5 V.

Further versions on request



Additional

- Position controller with standard signal input 4...20 mA or 0...10 V
- Feedback via potentiometer or 4...20 mA analogue signal
- Slowly rotating version
- 3-position actuator (90° or 180°)
- 3/2 way
- Emergency reset



Port connection

3-piece ball valves also available in versions with NPT thread or weld connection

5.4. Ordering chart accessories

Conversion sleeves

Description	Article no.
Conversion Sleeve star/square 14/9 mm	665288 
Conversion Sleeve star/square 14/11 mm	665289 
Conversion Sleeve square/square 17/14 mm	665290 
Conversion Sleeve star/square 17/14 mm	773348 
Conversion Sleeve star/square 17/11 mm	773343 
Conversion Sleeve square/square 22/19 mm	773836 
Conversion Sleeve star/square 22/17 mm	684858 
Conversion Sleeve star/star 22/14 mm	666684 
Conversion Sleeve star/square 22/11 mm	773344 
Conversion Sleeve star/square 27/22 mm	774594 
Conversion Sleeve square/square 27/19 mm	774279 
Conversion Sleeve square/square 27/17 mm	774193 

Replacement battery for actuators with electronic emergency reset function

Note:

- The battery has to be replaced every 2 years at the latest or after 500 charging cycles.
- Additional Info for Type 3004: The person who changes the battery must be trained and qualified to do modifications on ATEX products.
- Actuators which do not have an electronic emergency reset when delivered cannot be retrofitted with this.

Description	Article no.
Replacement battery for actuators of Type 3003 with electronic emergency reset function	770051 
Replacement battery for actuators of Type 3005 and 3004 (25Nm...300Nm) with electronic emergency reset function	772991 

Bürkert – Close to You

For up-to-date addresses
please visit us at
www.burkert.com

DTS 1000082140 EN Version: AA Status: RL (released | freigegeben | valide) printed: 06.08.2021





Electromagnetic flow sensor without flange (wafer type connection variant)

- For connection to a Type SE58 transmitter (with or without display, in compact or remote variant) for flow measurement
- Variant mainly for use in applications with water
- Flow measurement 25 to approx. 75.000 l/min for DN 25 to DN 400



Product variants described in the data sheet may differ from the product presentation and description.

Can be combined with



Type SE58 ▶
L variant of the transmitter for electromagnetic inductive flow sensors



Type SE58 ▶
M variant of the transmitter for electromagnetic-inductive flow sensors



Type SE58 ▶
S variant of the transmitter for electromagnetic-inductive flow sensors

Type description

The electromagnetic flow sensor Type S054 (compact or remote variant) is suitable for liquids with minimum conductivity and for use in applications with water measurement requirements.

Combining it with the corresponding transmitter Type SE58 S (required minimum conductivity: 20 μ S/cm) or Type SE58 M or Type SE58 L transmitters (required minimum conductivity: 5 μ S/cm) results in a flowmeter with varying performance, functions, materials, and approvals, suitable for specific applications as per the respective requirements.

Compact devices are built with Type SE58 S, while with Type SE58 M or Type SE58 L, both compact devices and remote variants are available, where the transmitter and sensor are connected with 2 cables up to a maximum distance. Standard process connections for Type S054 are available as wafer type connections.

Table of contents

1. General technical data	3
2. Approvals and conformities	5
2.1. Conformity	5
2.2. Standards	5
3. Materials	5
3.1. Bürkert resistApp	5
4. Dimensions	5
4.1. Compact variant	5
4.2. Remote variant with junction box	6
5. Product installation	6
5.1. Installation notes	6
Flow measurement	6
5.2. Selection of the nominal diameter	7
6. Product operation	7
6.1. Measuring principle	7
7. Ordering information	8
7.1. Bürkert eShop	8
7.2. Recommendation regarding product selection	8
7.3. Bürkert product filter	8
7.4. Ordering chart	8
7.5. Ordering chart accessories	9

1. General technical data

The electromagnetic flow sensor Type S054 in a compact or remote variant is intended for use with transmitter Type SE58, which is available in three variants L, M or S.



Further information can be found in the data sheet of the transmitter, see [data sheet Type SE58](#) ►.

Product properties

Material

Non-wetted parts

Sensor housing	Carbon steel painted (stainless steel 304 or 316 on request)
Junction box	Only for remote sensor: painted aluminium (on request: stainless steel 304 (1.4301) raw or polished)

Wetted parts

Lining	PP or ebonite (hard rubber) (PTFE on request)
Electrode	Stainless steel 316L (Alloy C276, Titanium, Tantalum, Platinum-rhodium on request)
Seal	- FKM (EPDM on request) with PP lining - Without gasket with ebonite (hard rubber) lining (with PTFE lining on request)
Pipe diameter	DN 25...DN 200 (upper DN on request)
Dimensions	Further information can be found in chapter "4. Dimensions" on page 5.
Measuring principle	Electromagnetic induction Further information can be found in chapter "6.1. Measuring principle" on page 7.
Measuring range	0...0.72 m³/h to 0...1130 m³/h (upper on request) Further information can be found in chapter "7.4. Ordering chart" on page 8.

Performance data

At reference conditions and according to internal test procedures:

- At room temperature
- Constant flow rate during the test, liquid speed > 1 m/s
- Pressure: > 30 Kpa
- Flow condition: observed inlet and outlet conditions
- Zero point stability: ± 0.005 %

Measurement deviation	If used with Type SE58 transmitter: • in compact or remote L variant: ≤ ± 0.2 % of the measured value for flow velocity > 0.5 m/s • in compact or remote M variant: ≤ ± 0.8 % of the measured value for flow velocity > 0.5 m/s • in compact S variant: ≤ ± 0.5 % of the measured value for flow velocity > 0.5 m/s See data sheet Type SE58 ►
Repeatability	If used with Type SE58 transmitter: • in compact or remote L variant: ≤ ± 0.1 % of the measured value for flow velocity > 0.5 m/s • in compact or remote M variant: ≤ ± 0.4 % of the measured value for flow velocity > 0.5 m/s • in compact S variant: ≤ ± 0.25 % of the measured value for flow velocity > 0.5 m/s See data sheet Type SE58 ►
Vacuum resistance	200 mbar (2.9 PSI) absolute at 100 °C (212 °F) for PTFE, at 60 °C (140 °F) for PP and at 80 °C (176 °F) for ebonite

Medium data

Fluid temperature	• With PP lining, if used with Type SE58 transmitter: – in compact variant: -0...+60 °C (+32...+140 °F) – in remote variant: -0...+60 °C (+32...+140 °F) • With ebonite lining, if used with Type SE58 transmitter: – in compact variant: -5...+80 °C (+23...+176 °F) – in remote variant: -5...+80 °C (+23...+176 °F) • With PTFE lining (on request), if used with Type SE58 transmitter: – in compact variant: -20...+100 °C (-4...+212 °F) – in remote variant: -20...+110 °C (-4...+230 °F)
Fluid pressure	• PN 16 (232 PSI) with PP or ebonite lining • PN 40 on request, only with PTFE lining up to DN 150
Minimum conductivity	5 µS/cm (or 20 µS/cm with demineralised water)

Product connections

Pipe connection	Wafer
Electrical connection	2 cable glands PG9 (for remote variant of the sensor)

Approvals and conformities

Directives

CE directive	Further information on the CE directive can be found in chapter “2.2. Standards” on page 5 .
Pressure equipment directive	The device is subject to the requirements of the Pressure Equipment Directive 2014/68/EU. Category II device for group 1 and 2 fluids under the following conditions: • maximum allowable pressure (PS) ≤ 40 bar • minimum/maximum temperature (TS): -10/+130 °C • within the following limits for liquids of group 2: – PN 10 for DN 400...DN 500 – PN 16 for DN 250...DN 300 – PN 25 for DN 200...DN 250 – PN 40 for DN 40...DN 250 • within the following limits for liquids of group 1 with a vapour pressure at the maximum allowable temperature not exceeding 0.5 bar (g): for diameters above DN 25 and PS x DN > 2000

Environment and installation

Ambient temperature	According to the used variant of Type SE58 transmitter and its material Further information can be found in the data sheet of the transmitter, see data sheet Type SE58 ▶.
Relative air humidity	≤ 90 %, without condensation
Height above sea level	Max. 2000 m
Operating condition	Continuous
Equipment mobility	Fixed
Application range	Indoor and outdoor Protect the device against electromagnetic interference, ultraviolet rays and, when installed outdoors, against the effects of climatic conditions.
Degree of protection according to IEC/EN 60529	If use with transmitter Type SE58: • in compact L and M variant: IP67 (IP68 optional) • in compact S variant: IP67 (IP68 optional) • in remote L and M variant: IP68
Installation category	Category II according to UL/EN 61010-1
Pollution degree	Degree 2 according to UL/EN 61010-1

2. Approvals and conformities

2.1. Conformity

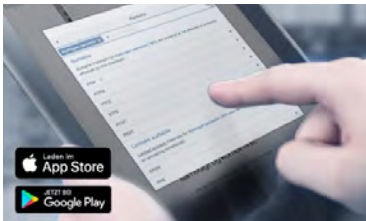
In accordance with the Declaration of Conformity, the product is compliant with the EU Directives.

2.2. Standards

The applied standards which are used to demonstrate compliance with the EU Directives are listed in the EU-Type Examination Certificate and/or the EU Declaration of Conformity.

3. Materials

3.1. Bürkert resistApp



Bürkert resistApp – Chemical resistance chart

You want to ensure the reliability and durability of the materials in your individual application case? Verify your combination of media and materials on our website or in our resistApp.

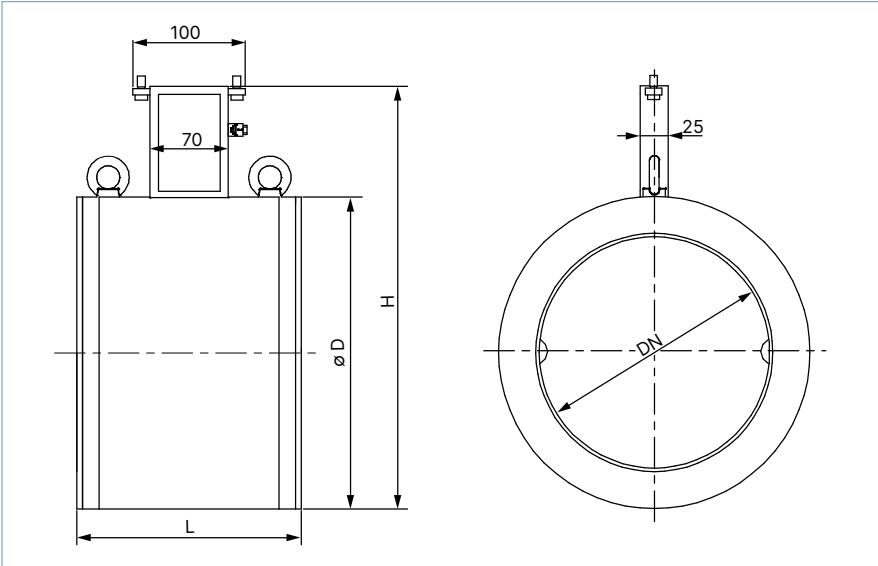
[Start chemical resistance check](#)

4. Dimensions

4.1. Compact variant

Note:

- Further information on the dimensions of the Type SE58 transmitter can be found in **data sheet Type SE58** ►.
- Dimensions in mm, unless otherwise stated



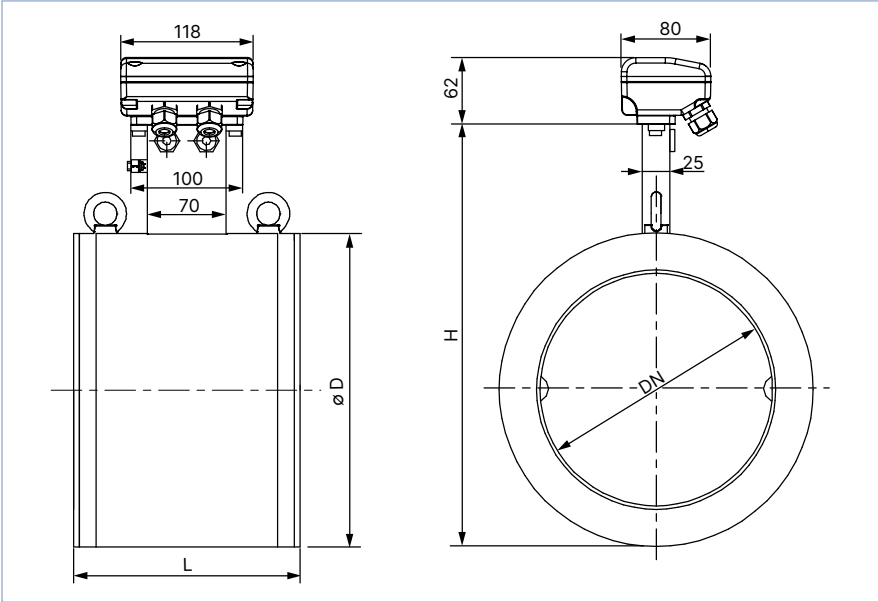
DN	H	L ¹⁾	Ø D
25	147	100	56
32	153	100	62
40	161	100	70
50	177	100	86
65	199	150	108
80	209	150	118
100	235	150	144
125	263	180	172
150	291	180	200
200	362	200	271

1.) tolerance +0/- 3 mm

4.2. Remote variant with junction box

Note:

- Further information on the dimensions of the Type SE58 transmitter can be found in **data sheet Type SE58** ▶.
- Dimensions in mm, unless otherwise stated



DN	H	L ¹⁾	Ø D
25	147	100	56
32	153	100	62
40	161	100	70
50	177	100	86
65	199	150	108
80	209	150	118
100	235	150	144
125	263	180	172
150	291	180	200
200	362	200	271

1.) Tolerance + 0/- 3 mm

5. Product installation

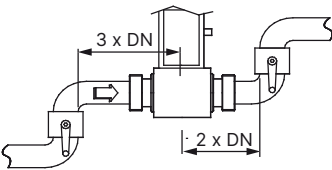
5.1. Installation notes

Flow measurement

Note:

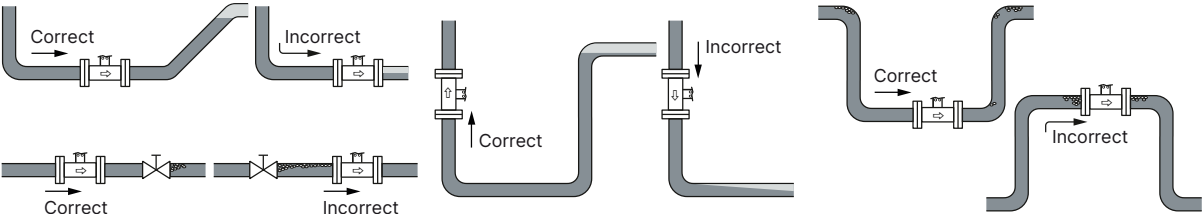
The device is not suitable for use in gaseous media and steam.

Minimum straight distances upstream and downstream of the sensor must be observed.



The device can be installed in either horizontal or vertical pipes, but following additional conditions should be respected:

- The pipe always must be filled with fluid at all times near the device, when it is in operation.
- Mount the sensor in the indicated positions shown below to obtain an accurate flow measurement.



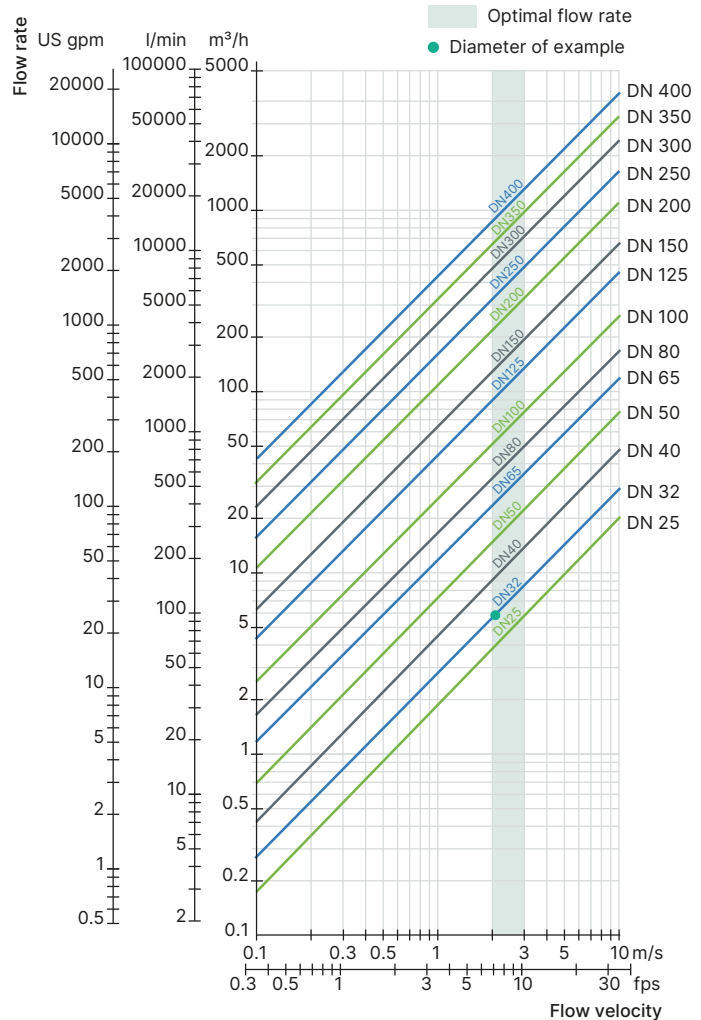
The suitable pipe size is selected using the diagram in the chapter "5.2. Selection of the nominal diameter" on page 7.

5.2. Selection of the nominal diameter

The following graph is used to determine the appropriate DN of the pipe and fitting for the application, according to the fluid velocity and the flow rate. On the chart, the intersection of flow velocity and flow rate gives the appropriate diameter.

Example:

- Nominal flow: 100 l/min
 - Optimal flow rate: 2...3 m/s
- Result: Select a pipe size of DN 32



6. Product operation

6.1. Measuring principle

Faraday's law serves as the physical basis for magnetic flow measurement.

Magnetic coils are arranged around the pipeline to generate a magnetic field. Conductive liquids flowing through the magnetic field induce a voltage at two opposite metallic electrodes in contact with the medium. These electrodes are used to measure the induced electrical alternating voltage.

The signal of sensor Type S054 must be amplified and processed by transmitter Type SE58.

Further information can be found in the data sheet of the transmitter, see **data sheet Type SE58** ►.

7. Ordering information

7.1. Bürkert eShop



Bürkert eShop – Easy ordering and quick delivery

You want to find your desired Bürkert product or spare part quickly and order directly? Our online shop is available for you 24/7. Sign up and enjoy all the benefits.

[Order online now](#)

7.2. Recommendation regarding product selection

A complete flowmeter consists of a Type S054 (compact or remote variant) and a Type SE58 transmitter (compact or remote variant).

See **data sheet Type SE58** ► for more information.

Two different components must be ordered to obtain a complete device. The following information is required:

- **Article no.** of the sensor **Type S054** (see **"7.4. Ordering chart" on page 8**)
- **Article no.** of the transmitter **Type SE58** (see **data sheet Type SE58** ► for more information)

7.3. Bürkert product filter



Bürkert product filter – Get quickly to the right product

You want to select products comfortably based on your technical requirements? Use the Bürkert product filter and find suitable articles for your application quickly and easily.

[Try out our product filter](#)

7.4. Ordering chart

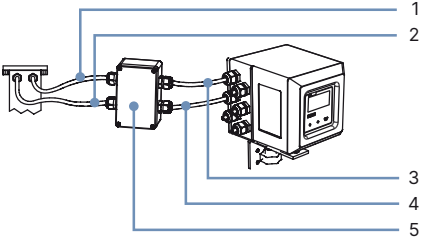
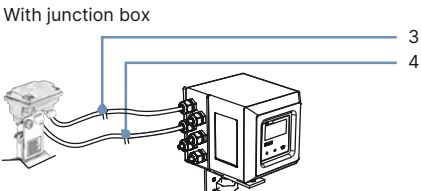
DN	Flow rate range		Pipe connection	Material				Article no.
[mm]	Min. 0...0.4 m/s	Max. 0...10 m/s		Housing	Electrode ¹⁾	Seal	Lining	
Sensor Type S054, compact variant								
25	0...0.72 m³/h	0...18 m³/h	Wafer type	Carbon steel	Stainless steel 316L	FKM	PP	554532 
32	0...1.16 m³/h	0...29 m³/h						559435 
40	0...1.80 m³/h	0...45 m³/h						554101 
50	0...2.88 m³/h	0...72 m³/h						554700 
65	0...4.80 m³/h	0...120 m³/h						559436 
80	0...7.20 m³/h	0...180 m³/h						554142 
100	0...11.20 m³/h	0...280 m³/h						554342 
125	0...18.00 m³/h	0...450 m³/h						562953 
150	0...25.60 m³/h	0...640 m³/h	Wafer type	Carbon steel	Stainless steel 316L	–	Ebonite (hard rubber)	562954 
200	0...45.20 m³/h	0...1130 m³/h						561912 

1.) 3 electrodes (2 measuring electrodes + 1 ground electrode)

Further variants on request			
	Material • Seal: EPDM • Lining: PTFE • Junction box: stainless steel 304 (1.4301) raw or polished • Body: stainless steel 304, stainless steel 316L • Electrodes: – Alloy C276 (2 measuring electrodes + 2 ground electrodes) – Titanium (2 measuring electrodes + 2 ground electrodes) – Tantalum (2 measuring electrodes + 2 ground electrodes) – Platinum-rhodium (2 measuring electrodes + 2 ground electrodes)		Orifice DN > 200 ¹⁾
			Pressure PN 10, PN 25, PN 40

1). Ebonite (hard rubber) or PTFE lining material (if PTFE not selected then ebonite as standard)

7.5. Ordering chart accessories

Accessories for remote sensor	No.	Description	Article no.
 Without junction box	1	10 m cable for electrodes ¹⁾ For connecting the sensor (variant without junction box) Type S051, S054, S055 or S056 to the connecting box of the extension cable set	448518 
	2	10 m cable for coils ¹⁾ For connecting the sensor (variant without junction box) Type S051, S054, S055 or S056 to the connecting box of the extension cable set	448519 
	3	10 m cable for electrodes ¹⁾ For connecting • the connecting box of the extension cable set to the transmitter Type SE58 • the sensor (variant with junction box) Type S051, S054, S055 or S056 to the transmitter Type SE58	562851 
 With junction box	4	10 m cable for coils ¹⁾ For connecting • the connecting box of the extension cable set to the transmitter Type SE58 • the sensor (variant with junction box) Type S051, S054, S055 or S056 to the transmitter Type SE58	562852 
	5	Connecting box of the extension cable set including resin	562853 

1). Cable lengths other than 10 m on request (for cables length > 20 m, a preamplifier is supplied for an additional charge).

DTS 1000471658 EN Version: D Status: RL (released | freigegeben | valide) printed: 09.05.2025



Transmitters for electromagnetic inductive flow sensors

- Must be combined with sensor Type S051, Type S054, Type S055 or Type S056
- Continuous measurement, high accuracy
- Different housing shapes and materials available
- Compact and remote design selectable, available with or without display

Product variants described in the data sheet may differ from the product presentation and description.

Can be combined with

	Type S051 ▶ Electromagnetic flow sensor, low flow rates
	Type S054 ▶ Electromagnetic flow sensor without flange (intermediate flange variant)
	Type S055 ▶ Electromagnetic sensor with flange
	Type S056 ▶ Electromagnetic flow sensor with hygienic process connections

Type description

The Type SE58 transmitter (in S, M or L variant) associated with the electromagnetic flow sensor (in compact or remote variant) Type S051, Type S054, Type S055 or Type S056 is designed for applications with a minimum conductivity of 5 µS/cm.

The variant S of the transmitter Type SE58 can only be used to build a compact flowmeter, with or without display. It is characterised by a housing with small external dimensions.

The variant without display has a housing and a cover in black painted aluminium or stainless steel.

The variant with display has a black painted aluminium or stainless steel housing and a plastic cover. The display is integrated in the cover. Flow rate and totalizer values are displayed simultaneously, as well as symbols for device status and alarms

The flow rate measurement can be transferred via a digital or an analogue output. The achievable uncertainty is 0.5 % of the measured value.

The device cannot be extended with the addition of input/output signals and functionalities such as dosing.

The variants M and L of the transmitter Type SE58, are available to build a compact or remote flowmeter, with or without display. Both variants are available with a housing made of painted aluminium, plastic or stainless steel.

The device can be extended with the addition of input/output signals and functionalities such as dosing (only Type SE58 L) and data logger. The Human Machine Interface (HMI) comprising the display and the operating keys allows choosing flexible data presentations, carrying out complete settings and configuring the device. The L variant is characterized by superior performance as shown by the low uncertainty of 0.2 % of the measured value compared to 0.4 % (option) and 0.8 % with the M variant.

Table of contents

1. General technical data	3
1.1. About the device.....	3
1.2. SE58 L transmitter.....	3
1.3. SE58 M transmitter.....	6
1.4. SE58 S transmitter.....	8
2. Approvals and conformities	9
2.1. Conformity	9
2.2. Standards.....	9
3. Dimensions	10
3.1. SE58 L and SE58 M transmitter.....	10
Compact variant with housing in aluminium or reinforced nylon	10
Remote variant with housing in aluminium or reinforced nylon.....	11
Compact and remote variants with housing in stainless steel	12
3.2. SE58 S transmitter.....	13
4. Performance specifications	14
4.1. Measurement deviation diagram	14
SE58 L transmitter	14
SE58 M transmitter.....	14
SE58 S transmitter.....	15
4.2. Default configuration.....	15
5. Product operation	16
6. Product accessories	17
7. Networking and combination with other Bürkert products	18
7.1. Compact variant.....	18
7.2. Remote variant	19
8. Ordering information	20
8.1. Bürkert eShop	20
8.2. Recommendation regarding product selection.....	20
8.3. Bürkert product filter	20
8.4. Ordering chart	21
SE58 L transmitter	21
SE58 M transmitter.....	22
SE58 S transmitter.....	23
8.5. Ordering chart accessories.....	23

1. General technical data

1.1. About the device

The transmitter Type SE58 is available in 3 variants:

- SE58 L available with an aluminium, plastic or stainless steel housing and cable glands, with or without display, compact or remote design
- SE58 M available with an aluminium, plastic or stainless steel housing and cable glands, with or without display, compact or remote design
- SE58 S available with an aluminium or stainless steel housing, M12 connector or one cable gland with connected cable and with or without display, no remote variant



Settings for SE58 L and SE58 M can be done using the operator keys or by USB cable and PC tool MCP. However, any changes using MCP are not recommended, unless they are:

- done after receiving corresponding training by Burkert,
- carried out by a professional,
- agreed by the end user, and
- done in accordance with the MCP manual.

All transmitters are intended for use with electromagnetic flow sensors Type S051, Type S054, Type S055 or Type S056.

Further information can be found in the data sheets of the electromagnetic-inductive flow sensors, see **data sheet Type S051** ▶, **data sheet Type S054** ▶, **data sheet Type S055** ▶ or **data sheet Type S056** ▶.

1.2. SE58 L transmitter



Product properties	
Material	
Lid	Polycarbonate (PC)
Front panel film	Polyester
Deckel	• Painted aluminium die casting or • Nylon reinforced (PA6) with 15 % glass fibre or
Housing	• Stainless steel 304 (1.4301), electro-polished • Painted aluminium die casting or • Nylon reinforced (PA6) with 15 % of glass fibre or • Stainless steel 304 (1.4301) electro-polished
Seal	Silicone
Cable gland	Polyamide (PA)
Display	Graphic display 8 lines x 16 characters, 128 × 64 pixels with back light
Keypad	3 operating keys
Compatibility	Electromagnetic flow sensors Type S051, Type S054, Type S055 or Type S056 in compact or remote variant Further information can be found in the data sheets, see data sheet Type S051 ▶, data sheet Type S054 ▶, data sheet Type S055 ▶ or data sheet Type S056 ▶.
Dimensions	Further information can be found in chapter “3. Dimensions” on page 10.

Data logger	A micro SD memory card 4 GB stores the selected data in a specified interval (option)
Special function	• Bidirectional measure • Dual measurement range • Diagnostic functions such as device self tests and process diagnostics like empty-pipe¹⁾ or measurement value limit detection • Batch filling functions
Performance data	
At reference conditions and according to internal test procedures:	
• At room temperature • Constant flow rate during the test, liquid speed > 1 m/s • Pressure: > 30 Kpa • Flow condition: observed inlet and outlet conditions • Zero point stability: $\pm 0.005\%$ • Default configuration if ordered together with sensor Type S051, Type S054, Type S055 or Type S056. Further information can be found in chapter "4.2. Default configuration" on page 15.	
Measurement deviation	$\leq \pm 0.2\%$ of the measured value for flow velocity > 0.5 m/s Further information can be found in chapter "4.1. Measurement deviation diagram" on page 14.
Repeatability	$\leq \pm 0.1\%$ of the measured value for flow velocity > 0.5 m/s
Response time	Minimum time • for analogue output (AO), when damping setting is deactivated and according to sensor size: – DN 03...DN 250: 20 ms – DN 300...DN 400: 100 ms • for digital output (DO): – 100 ms (if used with the sensor Type S054 or Type S055) – 20 ms (if used with the sensor Type S051 or Type S056)
Electrical data	
Operating voltage	• 100...240 V AC, 44 Hz...66 Hz • 12...48 V DC • Others on request
Power consumption	Max. 20 VA with 100...240 V AC operating voltage
Input	1 digital, function use is configurable (e.g. totalizer reset)
Output	• Transistor: – NPN or PNP (according to wiring), open collector – NPN output: 2 digital outputs (DO) – PNP output: only 1 digital output (DO) – configurable as – pulse/frequency (1250 Hz, max. 100 mA, 30 V DC) or – alarm/batch (adjustable usage) • Current: – max. 2 analogue outputs (AO), 0/4...20/22 mA, RL = 1000 Ω – HART (optional) only over first analogue output • Serial interface (optional): RS-485 (available with Modbus protocol (option))
Galvanic isolation	All the input/outputs are galvanically isolated up to 250 V from operating voltage
Protection class	Class I
Connection cable	Between sensor and transmitter: • cable C015/C016 for remote variant • 10 m (other lengths on request) Further information can be found in the data sheets, see data sheet Type S051 ▶, data sheet Type S054 ▶, data sheet Type S055 ▶ or data sheet Type S056 ▶.
Medium data	
Velocity range	0.4...10 m/s
Minimum conductivity	5 $\mu\text{S/cm}$
Product connections	
Electrical connection	• 5 cable glands PG11 for aluminium or nylon housing or • 6 cable glands PG11 for stainless steel housing

Plug for configuration connection	USB port for the connection to PC (USB cable with USB mini B and USB type A connectors is required for the configuration and parameter settings)
Industrial communication	
Supported network protocol	• Modbus RTU via RS-485 • HART (available on first analogue output)
Approvals and conformities	
Directives	
CE directive	Further information on the CE directive can be found in chapter "2.2. Standards" on page 9.
Environment and installation	
Ambient temperature	Operation and storage • Aluminium or stainless steel housing: - 20...+ 60 °C (- 4...+ 140 °F) • Reinforced nylon housing: - 10...+ 50 °C (+ 14...+ 122 °F)
Relative air humidity	0...100 %, without condensation
Height above sea level	- 200...+ 4000 m
Operating condition	Continuous
Equipment mobility	Fixed device
Application range	Indoor and outdoor Protect the device against electromagnetic interference, ultraviolet rays and, when installed outdoors, against the effects of climatic conditions.
Degree of protection according to IEC/EN 60529	• Aluminium housing: IP65, IP67 (IP68 option) • Reinforced nylon housing: IP65, IP67 • Stainless steel housing: IP65
Installation category	Category II according to UL/EN 61010-1
Pollution degree	Degree 2 according to UL/EN 61010-1

1.) Empty pipe functionality is not available if sensors are selected in the range of DN 03 to DN 20.

1.3. SE58 M transmitter



Product properties

Material

Lid	Polyamide (PA)
Front panel film	Polyester
Cover	- Painted aluminium die casting or - Nylon reinforced (PA6) with 15 % glass fibre or - Stainless steel 304 (1.4301), electro-polished
Housing	- Painted aluminium die casting or - Nylon reinforced (PA6) with 15 % glass fibre or - Stainless steel 304 (1.4301), electro-polished
Seal	Silicone
Cable gland	Polyamide (PA)
Display	Graphic display 8 lines x 16 characters, 128 × 64 pixels with back light
Keypad	3 operator keys
Compatibility	Electromagnetic flow sensors Type S051, Type S054, Type S055 or Type S056 in compact or remote variant Further information can be found in the data sheets, see data sheet Type S051 ▶, data sheet Type S054 ▶, data sheet Type S055 ▶ or data sheet Type S056 ▶.
Dimensions	Further information can be found in chapter "3. Dimensions" on page 10 .
Data logger	A micro SD memory card 4 GB stores the selected data in a specified interval (option)
Special function	- Bidirectional measure - Dual measurement range - Diagnostic functions such as device self tests and process diagnostics like empty-pipe¹⁾ or measurement value limit detection

Performance data

At reference conditions and according to internal test procedures:

- At room temperature
- Constant flow rate during the test, liquid speed > 1 m/s
- Pressure: > 30 Kpa
- Flow condition: observed inlet and outlet conditions
- Zero point stability: ± 0.005 %
- Default configuration if ordered together with sensor Type S051, Type S054, Type S055 or Type S056. Further information can be found in chapter ["4.2. Default configuration" on page 15](#).

Measurement deviation	$\leq \pm 0.8 \%$ of the measured value (optional: $\pm 0.4 \%$ of the measured value) for flow velocity > 0.5 m/s Further information can be found in chapter "4.1. Measurement deviation diagram" on page 14 .
Repeatability	$\leq \pm 0.4 \%$ of the measured value (optional: $\pm 0.2 \%$ of the measured value) for flow velocity > 0.5 m/s
Response time	Minimum time - for analogue output (AO), when damping setting is deactivated and according to sensor size: - DN 03...DN 250: 20 ms - DN 300...DN 400: 100 ms - for digital output (DO): 100 ms (if used with the sensor Type S054 or Type S055) 20 ms (if used with the sensor Type S051 or Type S056)

Electrical data

Operating voltage	100...240 V AC, 44 Hz...66 Hz 12...48 V DC - Others on request
Power consumption	Max. 20 VA with 100...240 V AC operating voltage
Input	1 digital, function use is configurable (e.g. totalizer reset)

Output	• Transistor: – NPN or PNP (according to wiring), open collector – NPN output: 2 digital outputs (DO) – PNP output: only 1 digital output (DO) – configurable as – pulse/frequency (1250 Hz, max. 100 mA, 30 V DC) or – alarm • Current: – max. 2 analogue outputs (AO), 0/4...20/22 mA, RL = 1000 Ω – HART (optional) only over first analogue output • Serial interface (optional): RS-485 (available with Modbus protocol (option))
Galvanic isolation	All the input/outputs are galvanically isolated up to 250 V from operating voltage
Protection class	Class I
Connection cable	Between sensor and transmitter: • cable C015/C016 for remote variant • 10 m (other lengths on request) Further information can be found in the data sheets, see data sheet Type S051 ▶, data sheet Type S054 ▶, data sheet Type S055 ▶, data sheet Type S056 ▶.
Medium data	
Velocity range	0.4...10 m/s
Minimum conductivity	5 µS/cm
Product connections	
Electrical connection	• 5 cable glands PG11 for aluminium or nylon housing or • 6 cable glands PG11 for stainless steel housing
Plug for configuration connection	USB port for the connection to PC (USB cable with USB mini B and USB type A connectors is required for the configuration and parameter settings)
Industrial communication	
Supported network protocol	• Modbus RTU via RS-485 • HART (available on first analogue output)
Approvals and conformities	
Directives	
CE directive	Further information on the CE directive can be found in chapter “2.2. Standards” on page 9 .
Environment and installation	
Ambient temperature	Operation and storage • Aluminium or stainless steel housing: - 20...+ 60 °C (- 4...+ 140 °F) • Reinforced nylon: - 10...+ 50 °C (+ 14...+ 122 °F)
Relative air humidity	0...100 %, without condensation
Height above sea level	- 200...+ 4000 m
Operating condition	Continuous
Equipment mobility	Fixed device
Application range	Indoor and outdoor Protect the device against electromagnetic interference, ultraviolet rays and, when installed outdoors, against the effects of climatic conditions.
Degree of protection according to IEC/EN 60529	• Aluminium housing: IP65, IP67 (IP68 optional) • Reinforced nylon housing: IP65, IP67 • Stainless steel housing: IP65
Installation category	Category II according to UL/EN 61010-1
Pollution degree	Degree 2 according to UL/EN 61010-1

1.) Empty pipe functionality is not available if sensors are selected in the range of DN 03 to DN 20.

1.4. SE58 S transmitter



Product properties

Material

Lid	Polyamide (PA)
Cover	PA6
Housing	- Painted Aluminium die casting or - Stainless steel AISI 304 (1.4301) raw or polished
Seal	NBR
Fixed connector	Nickel-plated brass
Cable gland	Polyamide (PA)
Display	LCD dimensions 60 × 40 mm, 2 lines + symbols, icons
Keypad	None
Compatibility	Electromagnetic flow sensors Type S051, Type S054, Type S055, Type S056 in compact variant, up to DN 400. Further information can be found in the data sheets, see data sheet Type S051 ►, data sheet Type S054 ►, data sheet Type S055 ►, data sheet Type S056 ►.
Dimensions	Further information can be found in chapter "3. Dimensions" on page 10 .
Parametrisation	Remote configuration by USB cable and PC tool MCP. However, any changes using MCP are not recommended, unless they are: - done after receiving corresponding training by Burkert, - carried out by a professional, - agreed by the end user, and - done in accordance with the MCP manual. Further information can be found in chapters "5. Product operation" on page 16 , "6. Product accessories" on page 17 and "8.5. Ordering chart accessories" on page 23 .
Data logger	An EEPROM stores the measured values (in case of power failure)
Special function	Bidirectional measure

Performance data

At reference conditions and according to internal test procedures:

- At room temperature
- Constant flow rate during the test, liquid speed > 1 m/s
- Pressure: > 30 Kpa
- Flow condition: observed inlet and outlet conditions
- Zero point stability: ± 0.005 %
- Default configuration if ordered together with sensor Type S051, Type S054, Type S055, Type S056. Further information can be found in chapter ["4.2. Default configuration" on page 15](#).

Measurement deviation	$\leq \pm 0.5\%$ of the measured value for flow velocity > 0.5 m/s Further information can be found in chapter "4.1. Measurement deviation diagram" on page 14 .
Repeatability	$\leq \pm 0.25\%$ of the measured value for flow velocity > 0.5 m/s
Response time	Minimum time - for analogue output (AO), when damping setting is deactivated and according to sensor size: - DN 03...DN 250: 20 ms - DN 300...DN 400: 100 ms - for digital output (DO): 100 ms (if used with the sensor Type S054 or Type S055) 20 ms (if used with the sensor Type S051 or Type S056)

Electrical data

Operating voltage	12...30 V DC if not using mA output 18...30 V DC if using mA output
Power consumption	Max. 1 W
Input	None

Output	- Transistor: - NPN only (PNP not possible), open collector - NPN output: 2 digital outputs (DO) - configurable as - pulse/frequency (1250 Hz, max. 100 mA, 30 V DC) or - alarms - Current: 1 analogue output (AO), 0/4...20 mA, RL = 500 Ω (18...30 V DC)
Protection class	Class I
Medium data	
Velocity range	0.4...10 m/s
Minimum conductivity	20 µS/cm
Product connections	
Electrical connection	1 × 5-pin M12 female connector (included in the delivery) or - Cable gland with 2 meter cable, already connected
Plug for configuration connection	USB port for the connection to PC (USB cable with USB mini B and USB type A connectors is required for the configuration and parameter settings)
Approvals and conformities	
Directives	
CE directive	Further information on the CE directive can be found in chapter "2.2. Standards" on page 9.
Environment and installation	
Ambient temperature	Operation and storage: - if analogue output used: - 20...+ 60 °C (- 4...+ 140 °F) - if no analogue output used: - 10...+ 60 °C (14...+ 140 °F)
Relative air humidity	0...100 %, without condensation
Height above sea level	- 200 up to + 2000 m
Operating condition	Continuous
Equipment mobility	Fixed device
Application range	Indoor and outdoor Protect the device against electromagnetic interference, ultraviolet rays and, when installed outdoors, against the effects of climatic conditions.
Degree of protection according to IEC/EN 60529	IP65, IP67 (IP68 optional)
Installation category	Category II according to UL/EN 61010-1
Pollution degree	Degree 2 according to UL/EN 61010-1

2. Approvals and conformities

2.1. Conformity

In accordance with the Declaration of Conformity, the product is compliant with the EU Directives.

2.2. Standards

The applied standards which are used to demonstrate compliance with the EU Directives are listed in the EU-Type Examination Certificate and/or the EU Declaration of Conformity.

3. Dimensions

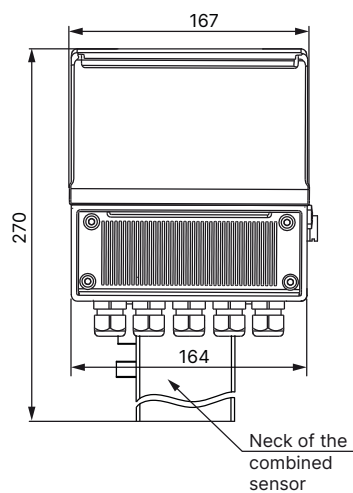
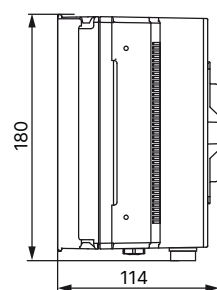
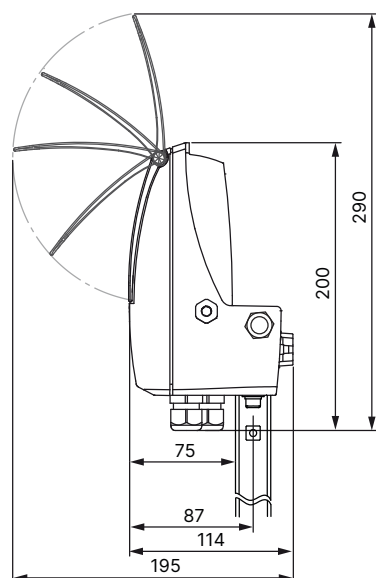
3.1. SE58 L and SE58 M transmitter

Compact variant with housing in aluminium or reinforced nylon

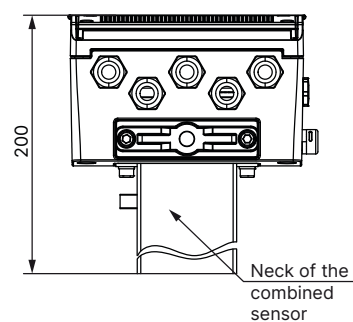
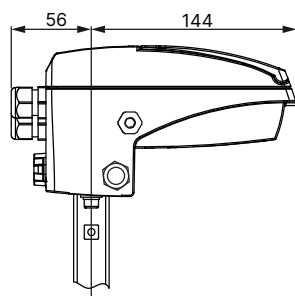
Note:

Dimensions in mm, unless otherwise stated

Compact variant, vertical mounting (standard)



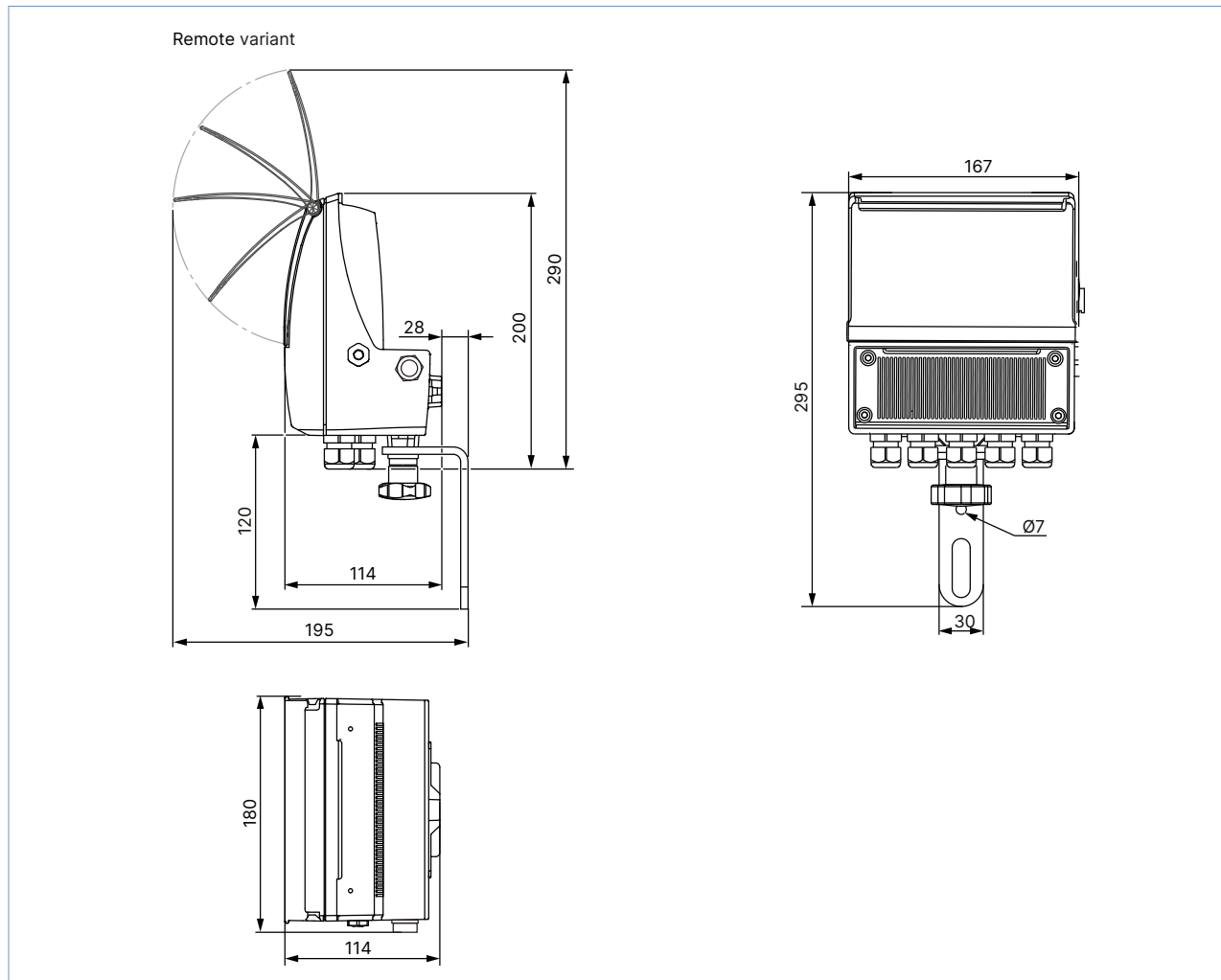
Compact variant, horizontal mounting (option)



Remote variant with housing in aluminium or reinforced nylon

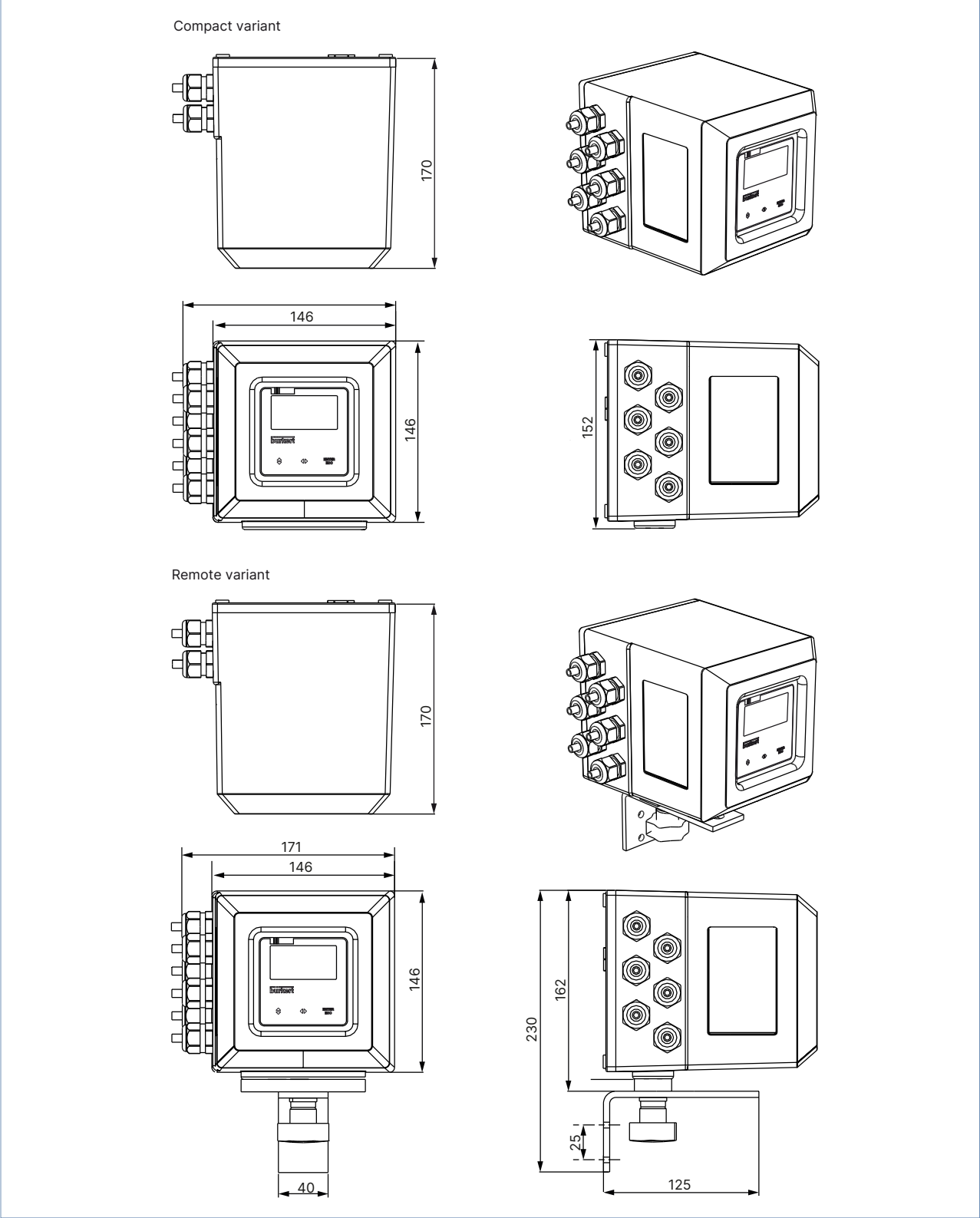
Note:

Dimensions in mm, unless otherwise stated



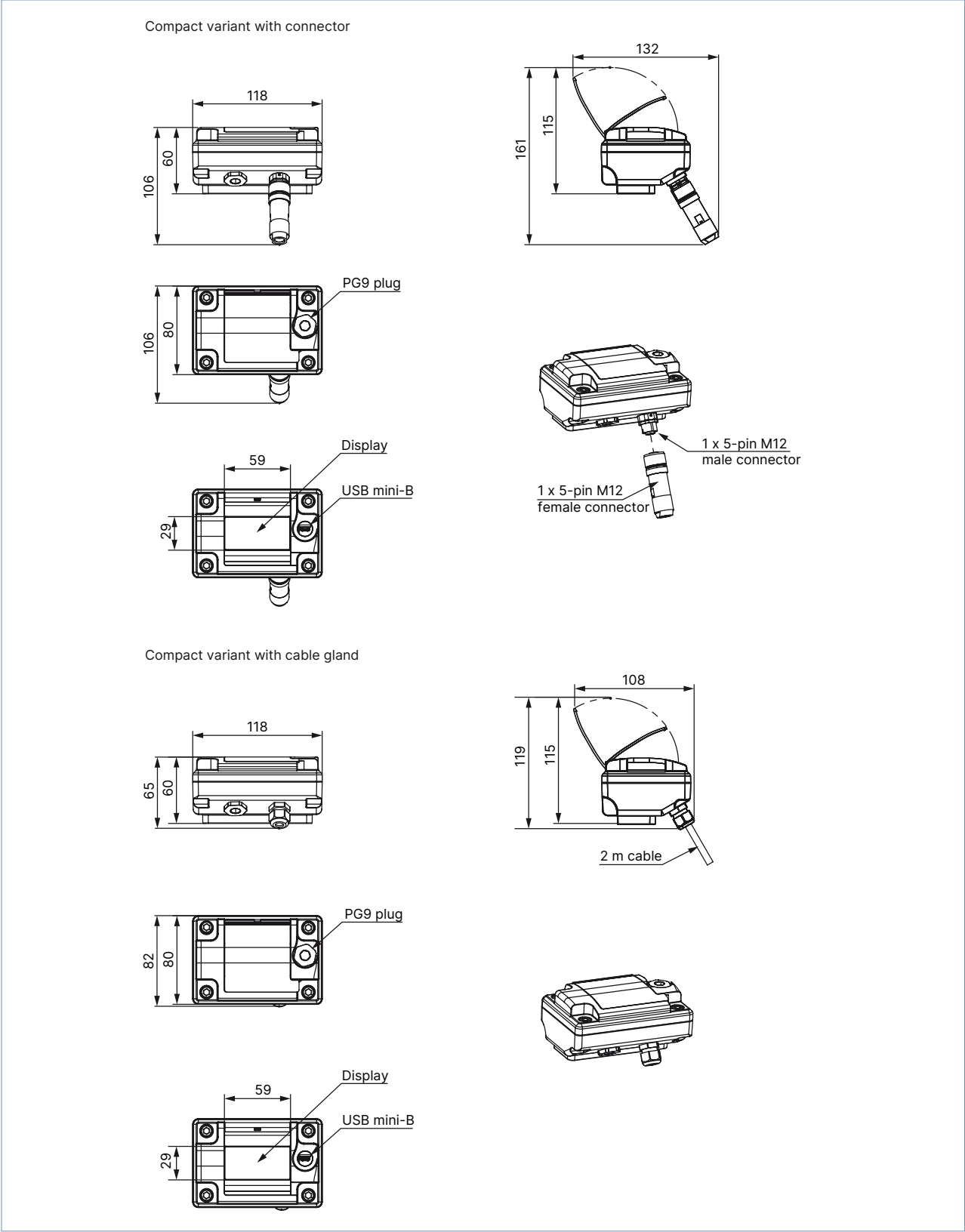
Compact and remote variants with housing in stainless steel

Note:
Dimensions in mm, unless otherwise stated



3.2. SE58 S transmitter

Note:
Dimensions in mm, unless otherwise stated



4. Performance specifications

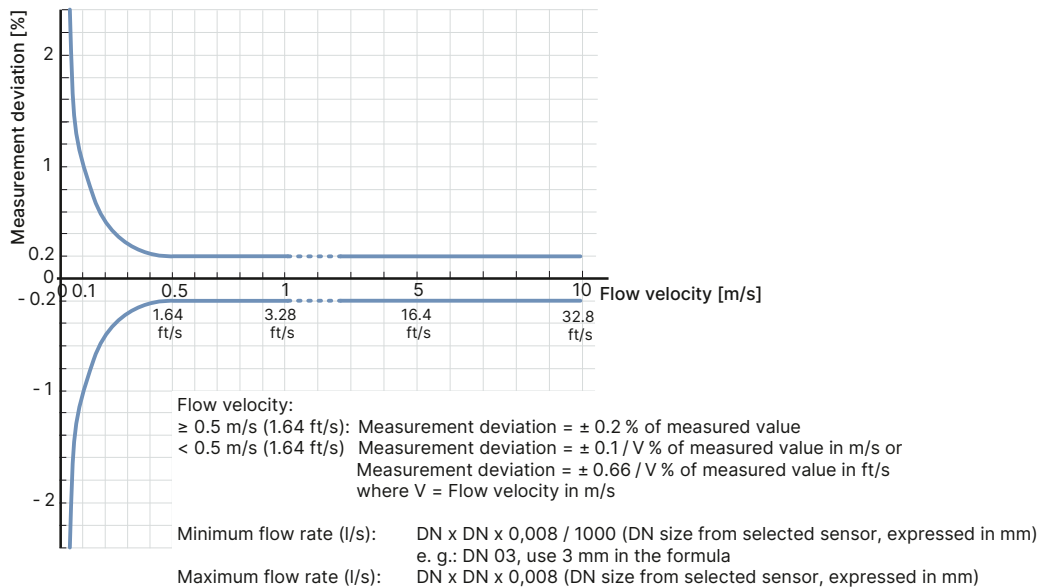
4.1. Measurement deviation diagram

SE58 L transmitter

Note:

This following diagram is valid for the complete device (transmitter Type SE58 L combined with a flow sensor Type S051, Type S054, Type S055 or Type S056).

See [data sheet Type S051](#) ▶, [Type S054](#) ▶, [Type S055](#) ▶ or [Type S056](#) ▶

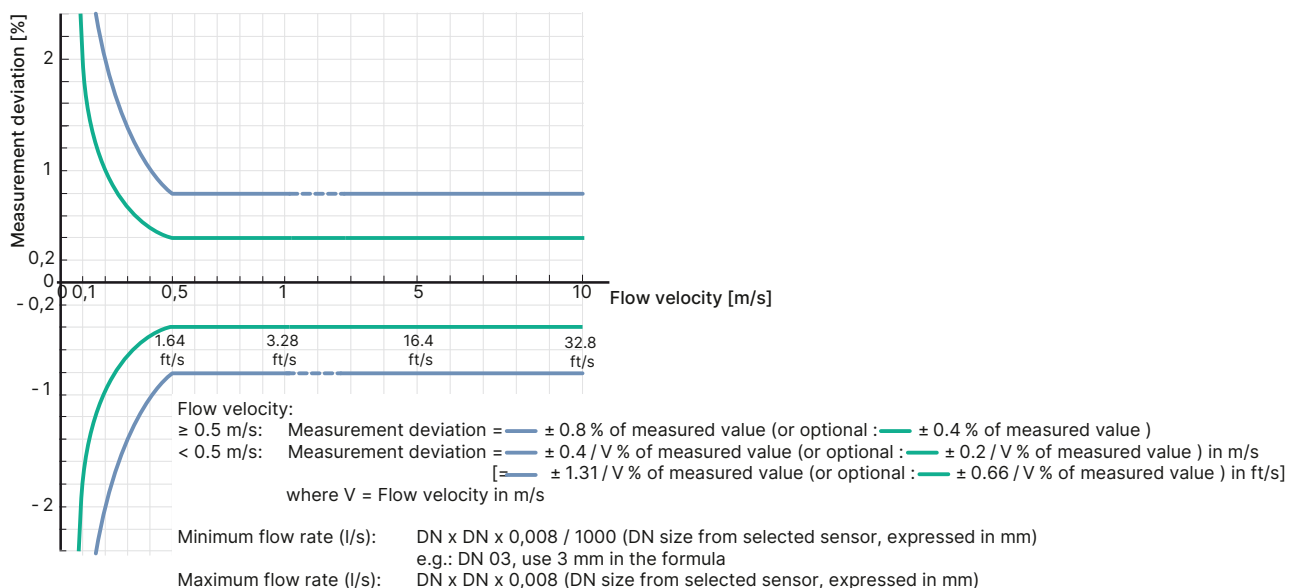


SE58 M transmitter

Note:

This following diagram is valid for the complete device (transmitter Type SE58 M combined with a flow sensor Type S051, Type S054, Type S055 or Type S056).

See [data sheet Type S051](#) ▶, [Type S054](#) ▶, [Type S055](#) ▶ or [Type S056](#) ▶

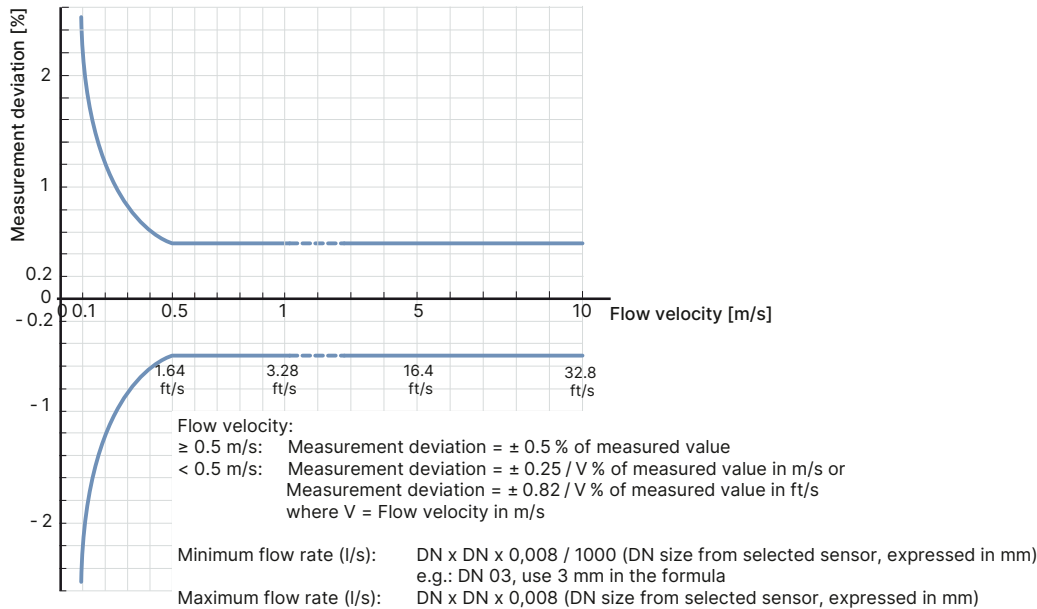


SE58 S transmitter

Note:

This following diagram is valid for the complete device (transmitter Type SE58 L and flow sensor Type S051, Type S054, Type S055 or Type S056).

See [data sheet Type S051](#) ▶, [Type S054](#) ▶, [Type S055](#) ▶ or [Type S056](#) ▶



4.2. Default configuration

Note:

This following diagram is valid for the complete device (transmitter Type SE58 L, Type SE58 M or Type SE58 S combined with a flow sensor Type S051, Type S054, Type S055 or Type S056).

See [data sheet Type S051](#) ▶, [Type S054](#) ▶, [Type S055](#) ▶ or [Type S056](#) ▶

DN	3	6	10	15	20	25	32	40	50	65	80	100	125	150	200	250	300	350	400	450	500
At 4 mA in m ³ /h	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
At 20 mA in m ³ /h	0.1	0.6	2	4	8	16	25	40	63	120	160	250	400	630	1000	1600	2500	2500	4000	4000	6300
Litre per pulse	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.5	0.5	1	2	10	10	10	10	10	10	10	100	100

5. Product operation

The configuration can be done in two different ways:

- by transmitter keypad if equipped with display (except for transmitter SE58 S)
- by USB cable and PC tool MCP (virtual display of instrument) for transmitter with or without display.

This MCP software runs under MS Windows and it is available for download on the Bürkert's website for free. However, any changes using MCP are not recommended, unless they are:

- done after receiving corresponding training by Bürkert,
- carried out by a professional,
- agreed by the end user, and
- done in accordance with the MCP manual which can be found under **Type SE58** ► on the Bürkert's website.

When using the MCP software, you agree to the following software Tools End User License Agreement "MCP" (STEULA):

Permission is hereby granted, free of charge, to any person obtaining a copy of this software and associated documentation files (the "Software") to use the software, and to permit persons to whom the software is furnished to do so subject to the conditions of this STEULA.

In a nutshell

- The software is intended for use by professionals and professionally, and in connection with our products only.
- Although we provide access to a convenient tool, it is not required for the use of our products. We are not liable for any consequence in using this software.
- We will update the software from time to time, but not regularly, and may at any point in time discontinue to offer the software or its updates for download.
- The software may include technical or other mistakes, inaccuracies or typographical errors.
- At any time without prior notice, we may make changes to the software pointing to third-party websites or shops or documentation made available on the third-party's website.
- The software may be out of date and we make no commitment to update such documents.

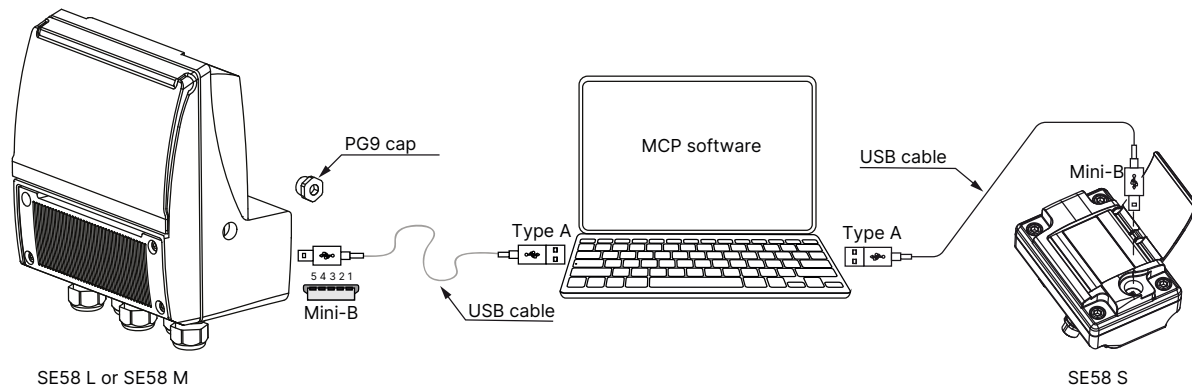
Legal advise

1. One registered copy of supplied software may either be used by a single person who uses the software personally on one or more computers, or installed on a single computer used non-simultaneously by multiple people, but not both.
2. You may access to software through a network, provided that you have obtained and agreed to individual licenses for the software. The licenses must cover all computers that will access the software through the network regardless if they access the software program simultaneously or at different times.
3. You are not allowed to modify its content, decompose, decompile its components, redistribute, offer or sell the software.
4. You are solely responsible for determining the appropriateness of using the software and assume any risks associated with your exercise of permissions under this license.
5. **This software and any accompanying files are given free of charge "as is" and without warranties, express or implied, as to performance or merchantability or non-infringement of third party rights.**
6. No advice or information, whether oral or written, obtained by you from us shall create any warranty for the software.
7. Good data processing procedure dictates that any program shall be thoroughly tested in a non-critical environment before using the software. You must assume the entire risk of using the program. **Note that using the software impacts the operability / functionality of the hardware and may have severe consequences for the production of the facility the hardware is installed in.**
8. **The software is in particular not designed, intended, licensed or authorized for use in any type of system or application in which the failure of the system or application could create a situation where personal injury or death may occur (e.g., medical systems, life support, life-sustaining systems, life-saving systems or security systems) or in hazardous environments requiring fail-safe controls, including without limitation, the design, construction, maintenance or operation of nuclear facilities, aircraft navigation or communication systems, air traffic control or weapons systems. The licensor specifically disclaims any express or implied warranty of fitness for such purposes.**
9. **In no event shall we be liable for any direct, indirect, incidental, special, exemplary or consequential damages (including, without limitation, procurement of substitute goods or services, loss of use, data or profits or business interruption) however caused and on any theory of liability, whether in contract, strict liability or tort (including negligence or otherwise) arising in any way out of the use of this software, even if advised of the possibility of such damage**

To download the MCP software, see **Type SE58** ► on the Bürkert's website.

6. Product accessories

An USB cable with USB mini B and USB type A connectors serves as the interface between computer and transmitter (see [“8.5. Ordering chart accessories” on page 23](#) and [“5. Product operation” on page 16](#)).



7. **Networking and combination with other Bürkert products**

7.1. **Compact variant**

Note:

The compact Type SE58 transmitter is intended for use with Type S051, Type S045, Type S055 or Type S056 compact flow sensors.

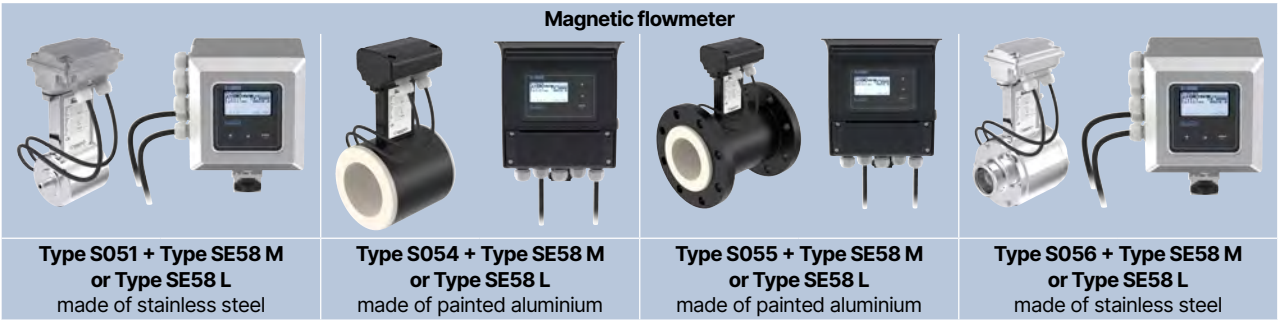
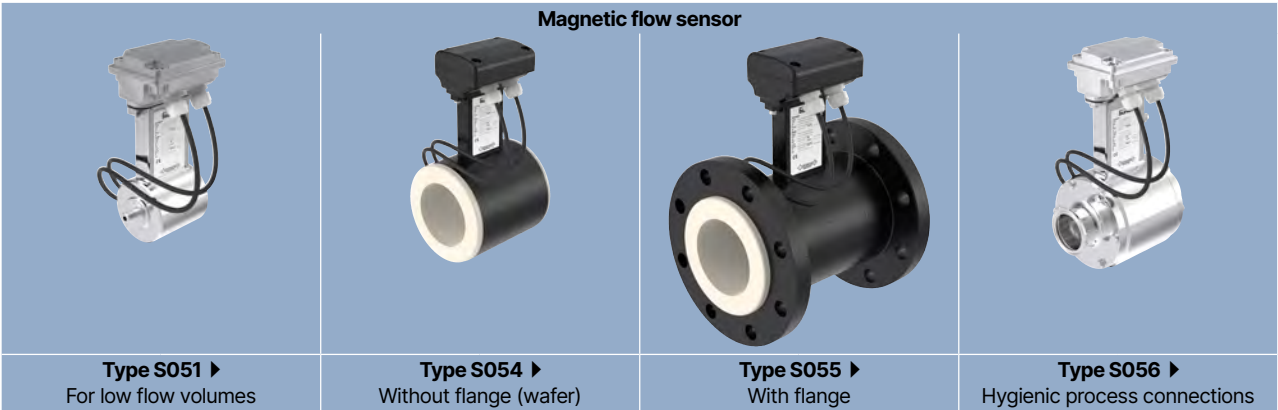
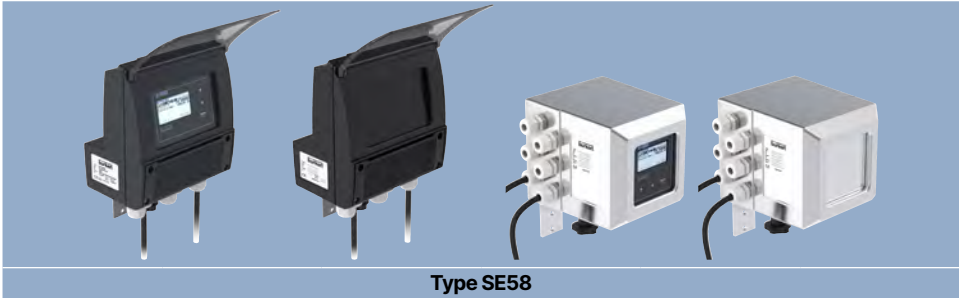
Example:



7.2. Remote variant

Note:
The Type SE58 transmitter is intended for use with Type S051, Type S045, Type S055 or Type S056 flow sensors, each in design for the remote variant.

Example:



8. Ordering information

8.1. Bürkert eShop



Bürkert eShop – Easy ordering and quick delivery

You want to find your desired Bürkert product or spare part quickly and order directly? Our online shop is available for you 24/7. Sign up and enjoy all the benefits.

[Order online now](#)

8.2. Recommendation regarding product selection

A complete full bore flowmeter consists of a flow sensor (compact or remote variant) Type S051, Type S054, Type S055 or Type S056 and a flow transmitter (compact or remote variant) Type SE58.

See [Data sheet Type S051](#) ▶, [Type S054](#) ▶, [Type S055](#) ▶ or [Type S056](#) ▶ for more information.

Two different components must be ordered to obtain a complete device. The following information is required:

- **Article no.** of the sensor [Type S051](#), [Type S054](#), [Type S055](#) or [Type S056](#) (see [Data sheet Type S051](#) ▶, [Type S054](#) ▶, [Type S055](#) ▶ or [Type S056](#) ▶ for more information.)
- **Article no.** of the transmitter [Type SE58](#) (see chapter [“SE58 L transmitter” on page 21](#), [“SE58 M transmitter” on page 22](#) or [“SE58 S transmitter” on page 23](#).)

8.3. Bürkert product filter



Bürkert product filter – Get quickly to the right product

You want to select products comfortably based on your technical requirements? Use the Bürkert product filter and find suitable articles for your application quickly and easily.

[Try out our product filter](#)

8.4. Ordering chart

SE58 L transmitter

Note:

- Not all SE58 L transmitter variants are listed in the following table. If the desired variant is not mentioned, please contact your Bürkert representative.
- The following variants are vertically mounted variants.

Operating voltage	Output	Network protocol	Housing material	Electrical connection	Article no.
Compact variant with display					
100...240 V AC	2 digital outputs (transistors)	Without	Aluminium	5 cable glands	571500 
			Stainless steel	6 cable glands	571507 
	2 digital outputs (transistors) + analogue output (4...20 mA)		Aluminium	5 cable glands	571501 
			Stainless steel	6 cable glands	571508 
12...48 V DC	2 digital outputs (transistors) + analogue output (4...20 mA)		Aluminium	5 cable glands	571502 
			Stainless steel	6 cable glands	571509 
Remote variant ¹⁾ (wall-mounting) with display					
100...240 V AC	2 digital outputs (transistors)	Without	Aluminium	5 cable glands	571505 
			Stainless steel	6 cable glands	571510 
	2 digital outputs (transistors) + analogue output (4...20 mA)		Aluminium	5 cable glands	571506 
			Stainless steel	6 cable glands	571511 
12...48 V DC	2 digital outputs (transistors) + analogue output (4...20 mA)		Aluminium	5 cable glands	571503 
			Stainless steel	6 cable glands	571513 

1.) Remote variants include two 10 m signal cable. If a longer cable is needed please consider that for ordering.

Further variants on request



Material

Nylon reinforced



Additional

- Compact variant for horizontal mounting or remote variant
- Variant with display (visible from the top or from the front) or without display
- Outputs:
 - 4...20 mA (1 or 2)
 - RS 485 (with protocol Modbus)
 - 2 transistors
- Industrial communication:
 - RS 485 (with protocol Modbus)
 - HART
 - Wi-Fi (for parameter settings)
- Data logger with memory card 4GB
- Degree of protection according to IEC/EN 60529: IP68

SE58 M transmitter

Note:

- Not all SE58 M transmitter variants are listed in the following table. If the desired variant is not mentioned, please contact your Bürkert representative.
- The following variants are vertically mounted variants and delivered with a measurement deviation of 0.8 % from the measured value.

Operating voltage	Outputs	Industrial communication	Housing material	Electrical connection	Article no.
Compact variant with display					
100...240 V AC	2 digital outputs (transistors)	Without	Nylon reinforced	5 cable glands	571540 
			Stainless steel	6 cable glands	571548 
	2 digital outputs (transistors) + analogue output (4...20 mA)		Nylon reinforced	5 cable glands	571541 
			Stainless steel	6 cable glands	571549 
12...48 V DC	2 digital outputs (transistors)		Nylon reinforced	5 cable glands	571542 
			Stainless steel	6 cable glands	571550 
	2 digital outputs (transistors) + analogue output (4...20 mA)		Nylon reinforced	5 cable glands	571543 
			Stainless steel	6 cable glands	571551 
Compact variant without display					
100...240 V AC	2 digital outputs (transistors)	Without	Nylon reinforced	5 cable glands	571544 
			Stainless steel	6 cable glands	571552 
	2 digital outputs (transistors) + analogue output (4...20 mA)		Nylon reinforced	5 cable glands	571545 
			Stainless steel	6 cable glands	571553 
12...48 V DC	2 digital outputs (transistors)		Nylon reinforced	5 cable glands	571546 
			Stainless steel	6 cable glands	571554 
	2 digital outputs (transistors) + analogue output (4...20 mA)		Nylon reinforced	5 cable glands	571547 
			Stainless steel	6 cable glands	571555 

Further variants on request



Material
Aluminium

**Additional**

- Compact variant for horizontal mounting or remote variant
- Variant with display (visible from the top or from the front) or without display
- Outputs :
 - 4...20 mA (1 or 2)
 - RS 485 (with protocol Modbus)
 - 2 transistors
- Measurement deviation of 0.4 % of the measured value
- Industrial communication:
 - RS 485 (with protocol Modbus)
 - HART
 - Wi-Fi (for parameter settings)
- Data logger with memory card 4GB
- Degree of protection according to IEC/EN 60529: IP68

SE58 S transmitter

Note:

- Not all SE58 S transmitter variants are listed in the following table. If the desired variant is not mentioned, please contact your Bürkert representative.

Operating voltage	Outputs	Industrial communication	Housing material	Electrical connection	Article no.
Compact variant with display					
12...30 V DC ¹⁾	2 digital outputs (transistors)	Without	Stainless steel polished	Cable gland with 2 m cable, already connected	571582 
	2 digital outputs (transistors) + analogue output (4...20 mA)				571583 
Compact variant without display					
12...30 V DC ¹⁾	2 digital outputs (transistors)	Without	Stainless steel polished	Cable gland with 2 m cable, already connected	571580 
	2 digital outputs (transistors) + analogue output (4...20 mA)				571581 

1.) 12...30 V DC if not using mA output or 18...30 V DC if using mA output

Further variants on request	
	Electrical connection 1 × 5-pin M12 female connector
	Material Aluminium, raw stainless steel, polished stainless steel
	Additional • Variant with display • Degree of protection according to IEC/EN 60529: IP68

8.5. Ordering chart accessories

Description	Article no.
USB cable with mini B and A connectors, cable length: 1.8 m	919499